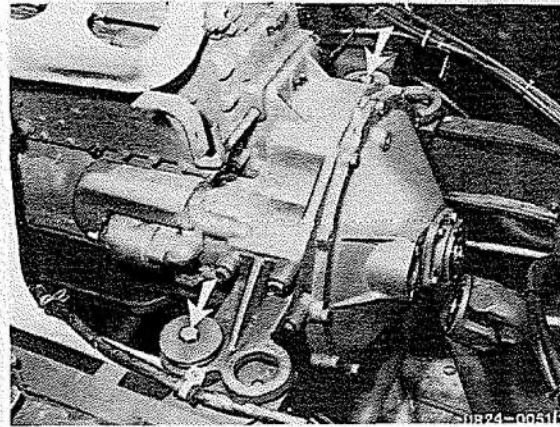
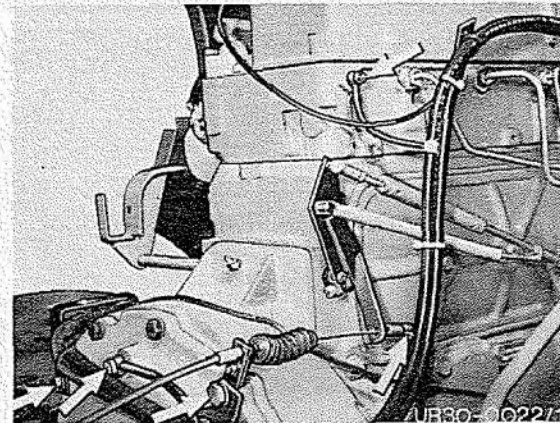


3 Locate and tighten fastening bolts of rear engine mounts.
See 1.4/1 for tightening torque.

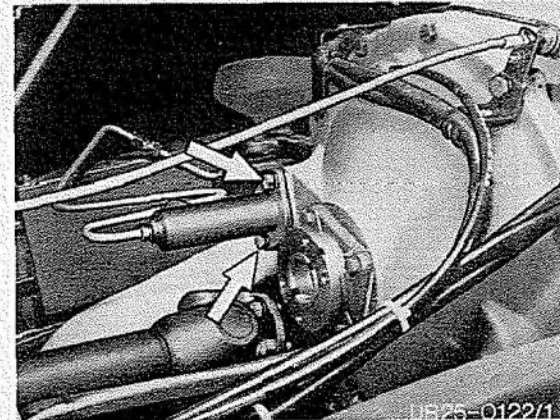
4 Lower engine and remove special tool No. 19.
Attach engine breather filter with sealant No. 2.



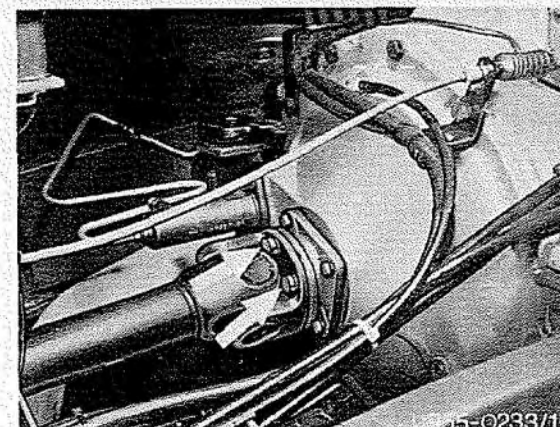
5 Remove starter wiring loom and venting line from clutch housing. Also remove fuel line, leak-off line, cold start line and idling control cable as fitted to special version 35 989.
Adjust idling setting, see 30.4-2.1/1.

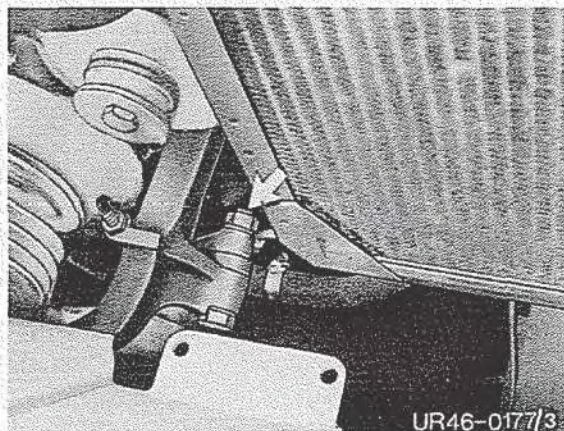


6 Attach clutch slave cylinder after coating sealing surface with sealant No. 2.

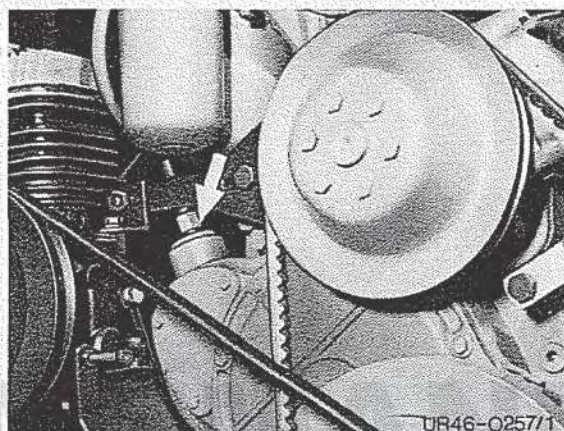


7 Attach propeller shaft to input flange.

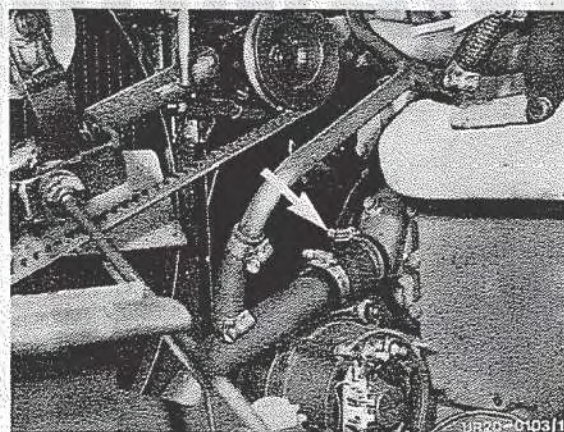




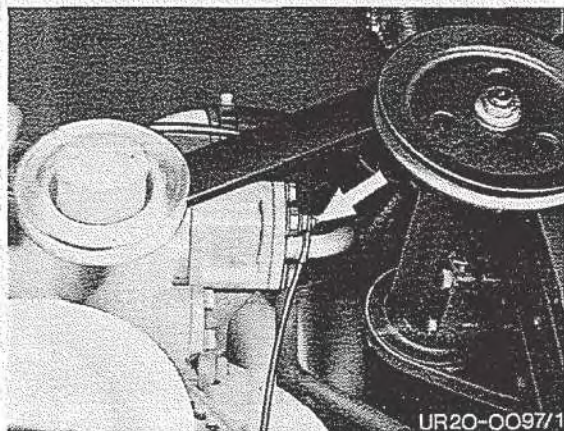
- 8 Attach hydraulic pump and bracket to engine support.



- 9 Attach venting line to timing gear case.

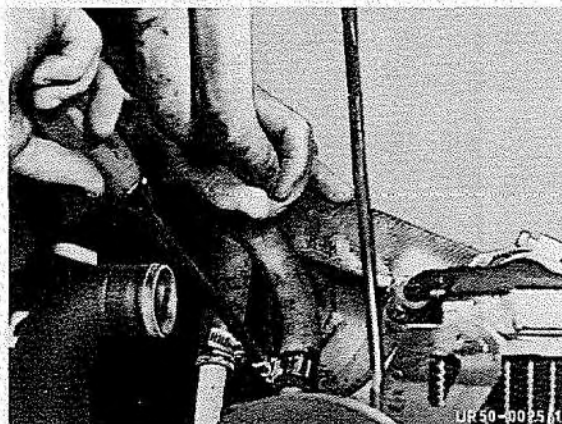


- 10 Attach coolant pipe to coolant pump and reservoir.

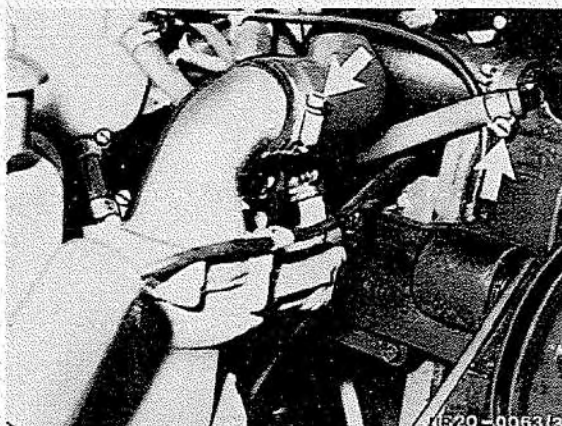


- 11 Attach electric line to coolant temperature pick-up.

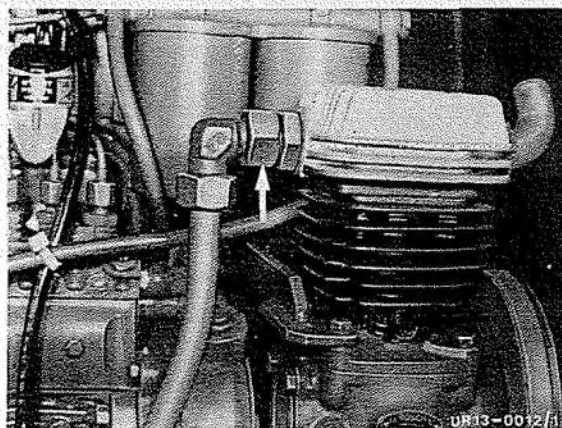
12 Connect heater hose.



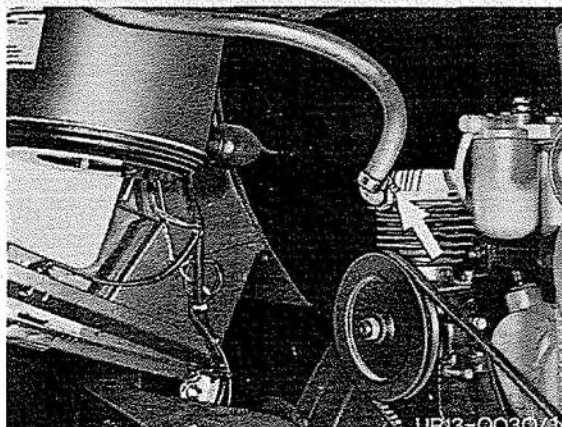
13 Attach coolant hose to thermostat housing and coolant venting hose to radiator.



14 Connect delivery line to compressor.

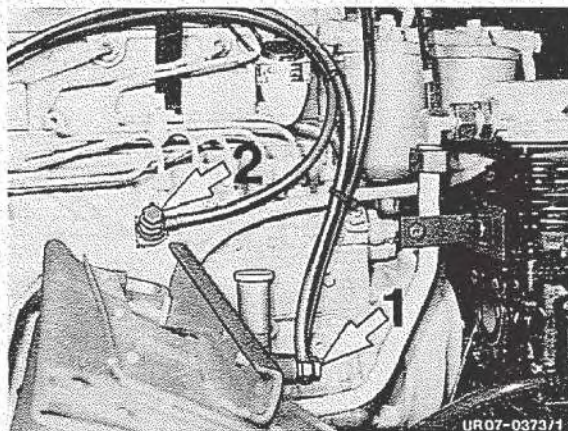


15 Attach intake line to compressor.



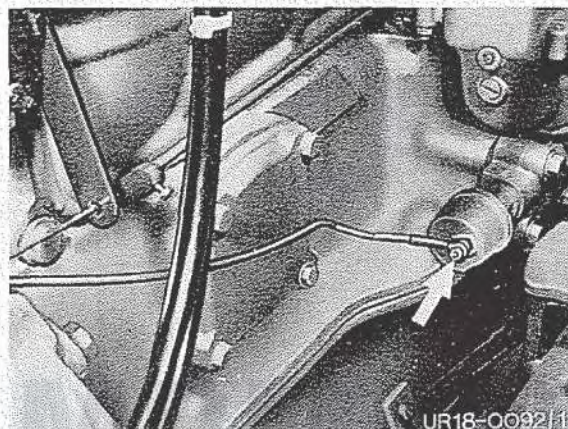
01.8 Removal and Installation of Engine

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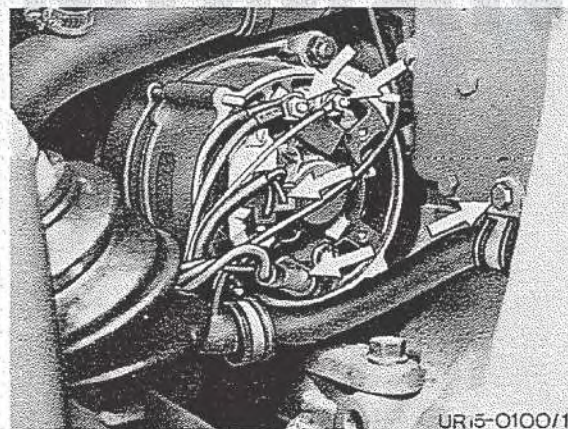


16 Attach fuel intake (1) and return lines (2).

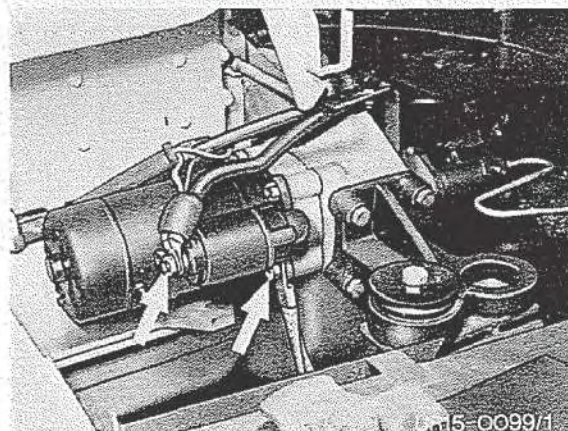
Note: The vent on the fuel filter is no longer fitted from chassis end No. 086 284.



17 Connect electrical line to oil pressure pickup.



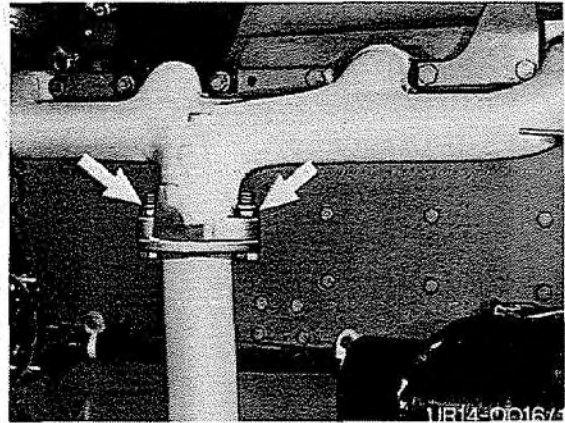
18 Referring to marks, attach electrical lines to generator and wiring harness to oil cooler.



19 Attach electrical lines and ground strap to starter.

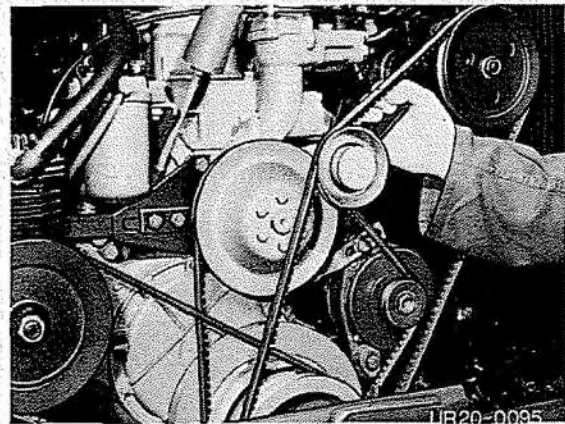
- 20 Attach exhaust pipe to exhaust manifold.
See 1.4/1 for tightening torque.

Note: For engine model 353.959, attach exhaust pipe to turbocharger.



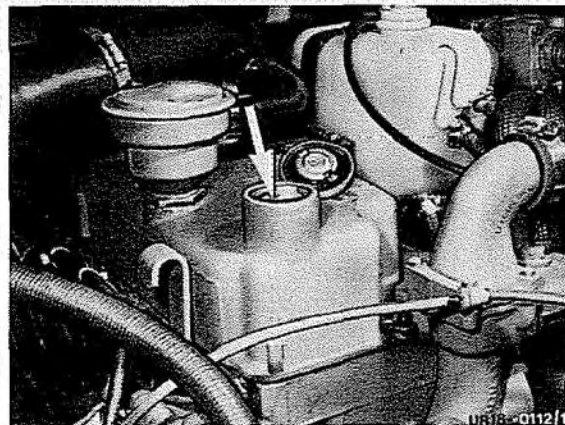
- 21 Attach V-belt for intermediate fan shaft.

- 22 Connect battery.



- 23 Lower cab,
referring to 60.5–2.1/1 or 60.5–2.2/1.

- 24 Fill with engine oil as required.
See 1.3/2 for quantity.

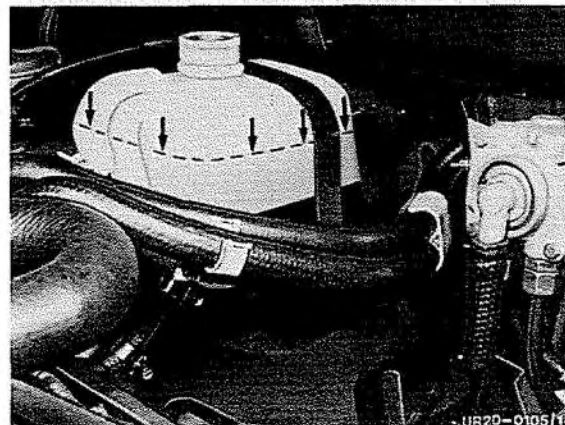


- 25 Vent fuel system, referring to 07.8–6.1/1.

- 26 Fill with coolant. Check cooling system for
leakage, referring to 20.8–5.1/1.
See 1.3/2 for quantity.

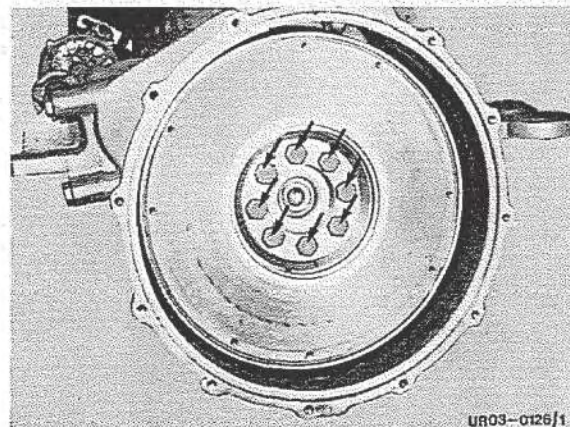
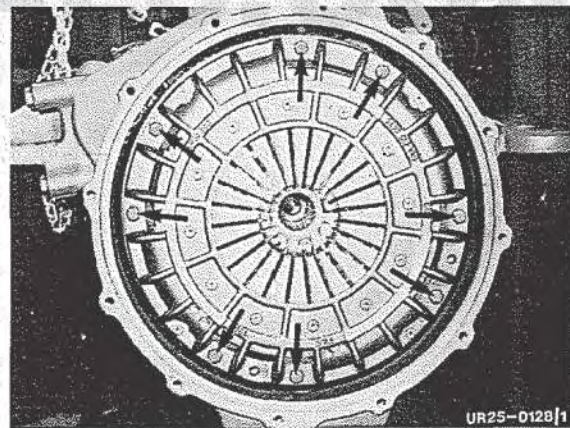
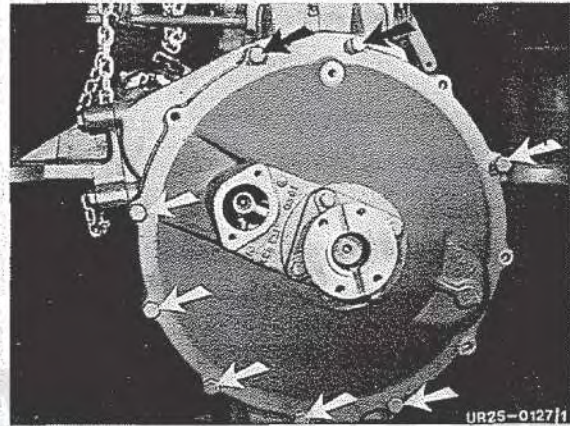
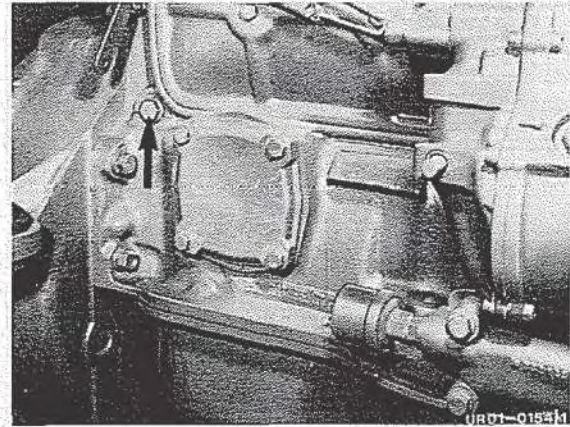
Note: The reservoir ought to be 2 thirds full.

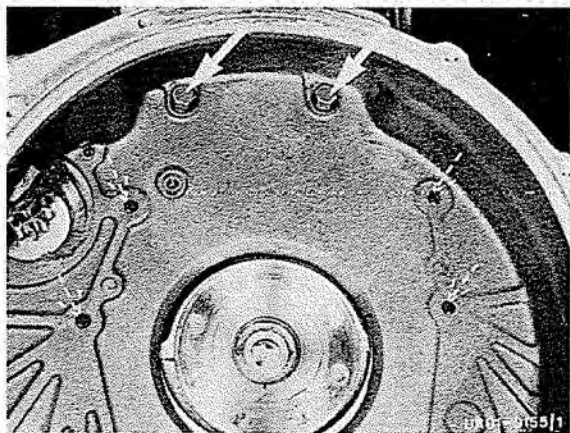
- 27 Carry out trial run.



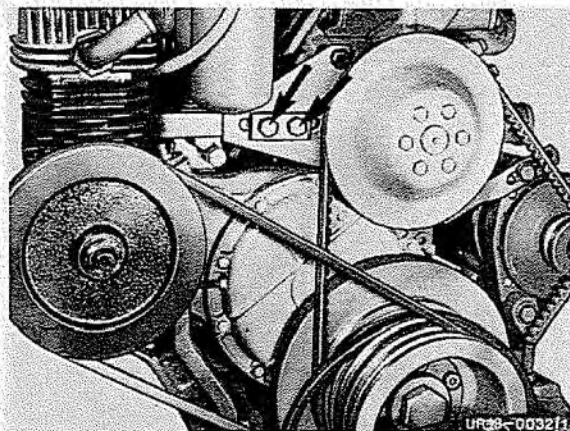
Removal

- 1 Remove engine, referring to 2.1/1.
- 2 Drain coolant at crankcase.
- 3 Detach clutch casing and input shaft.
- 4 Detach and remove clutch.
- 5 Detach flywheel.

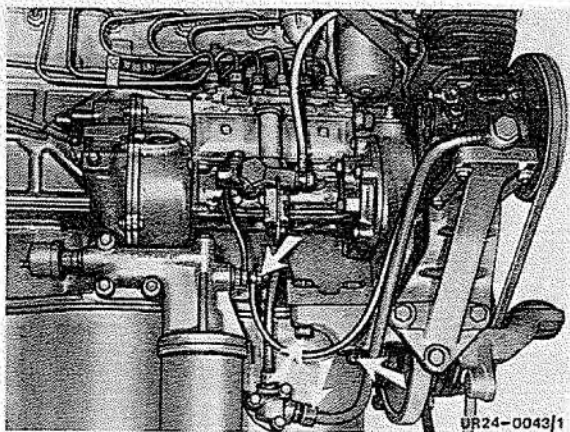




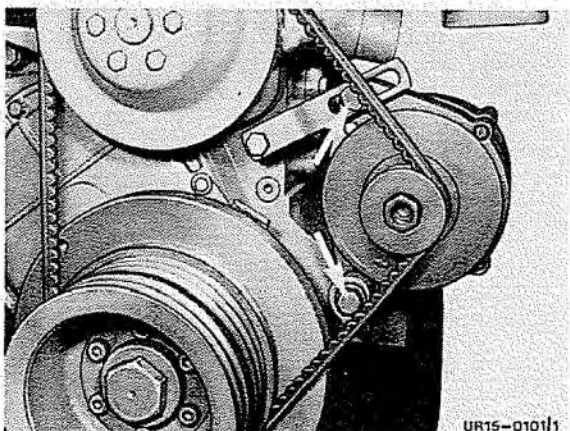
6 Detach flywheel housing.



7 Unscrew air compressor at bracket.

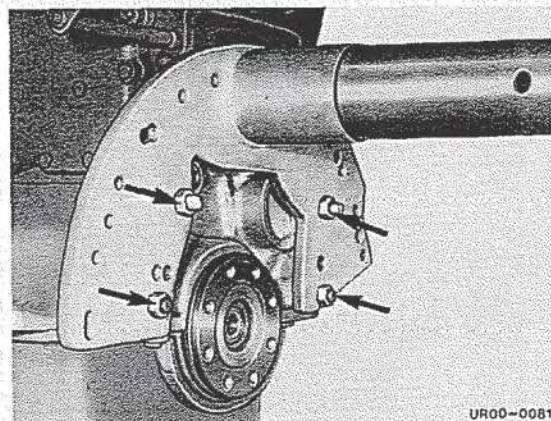


8 Detach air compressor lube lines and front engine support.

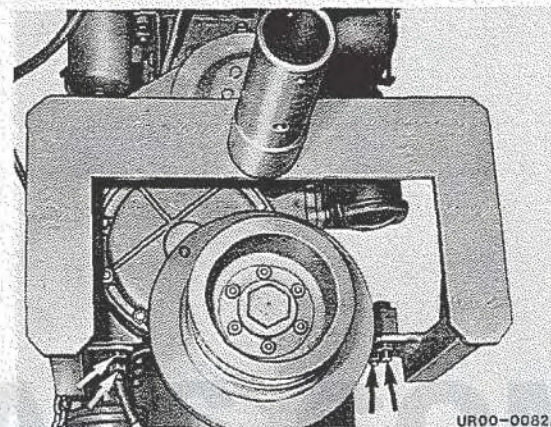


9 Detach generator.

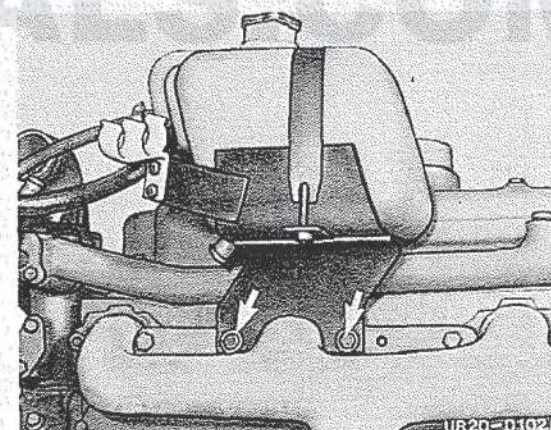
- 10 Attach assembling fixture to rear end of crank-case.



- 11 Attach assembling fixture to front end of crank-case. Position engine in trestle.

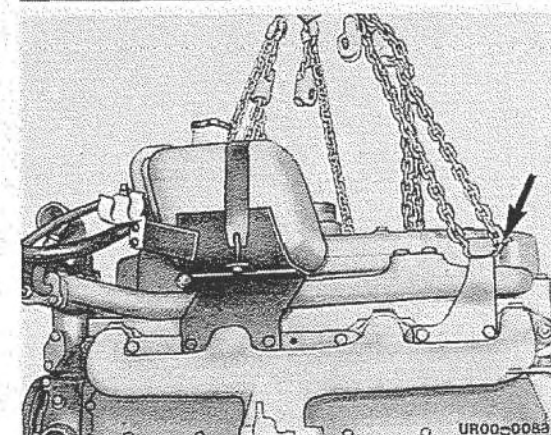


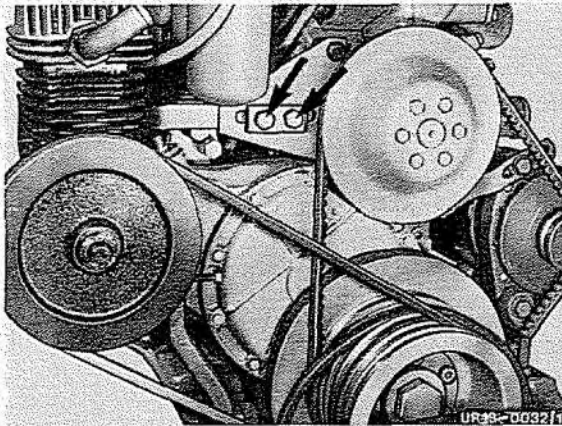
- 12 Detach coolant reservoir.



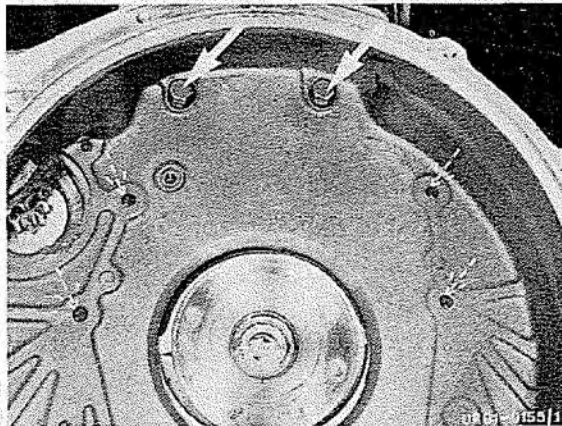
Installation

- 1 Attach and tighten coolant reservoir.
- 2 Attach engine to special tool No. 19 and remove assembling fixture at front and rear.
- 3 Attach generator.
- 4 Attach and tighten front engine support, referring to 1.4/1.

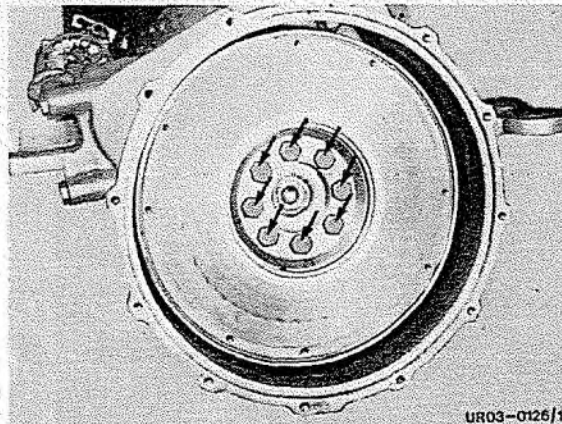




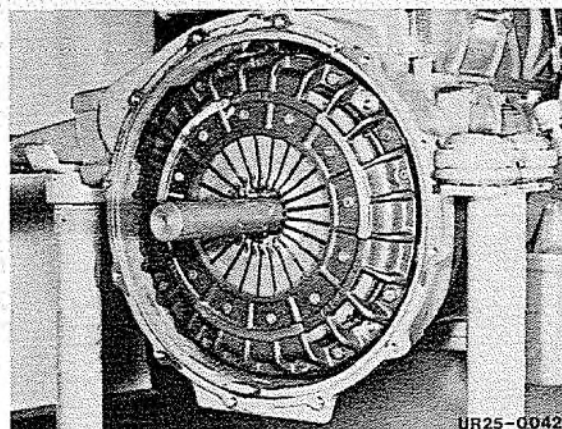
5 Attach lube lines to air compressor and air compressor to bracket. Tighten with reference to 1.4/1 for tightening torque.



6 Coat flywheel housing with sealant No. 3 apply sealant No. 4 to bolts and then tighten. See 1.4/1 for tightening torque.



7 Attach and tighten flywheel, referring to 1.4/1 for tightening torque.



8 Using special tool No. 6 attach and tighten clutch. See 25.10—1.4/1 for tightening torque.

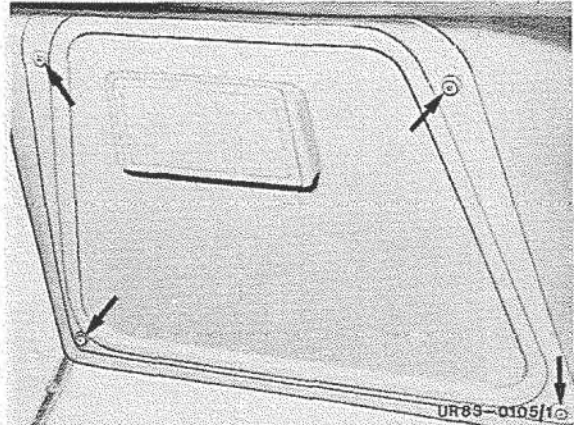
9 Coat clutch housing with sealant No. 2, apply locking fluid No. 1 to bolts and then tighten. See 25.10—1.4/1 for tightening torque.

10 Install engine, referring to 2.1/1.

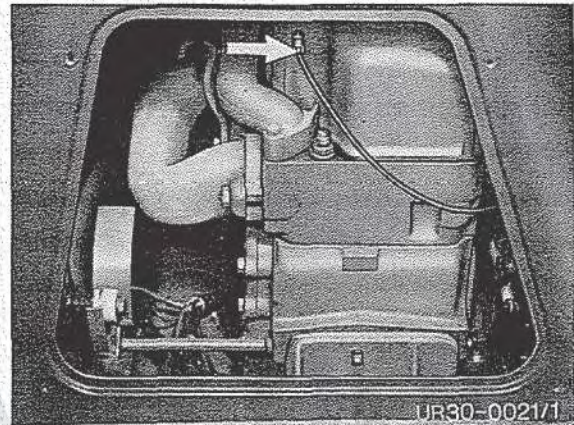
Removal

Note: Removal is similar for both engine models No. 353.959 and 353.961.

- 1 Unscrew engine hood at inside.

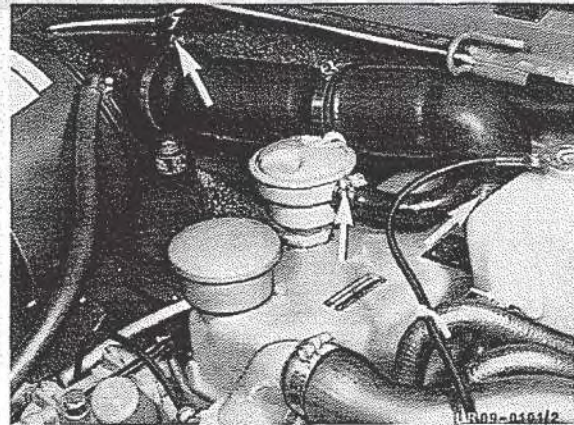


- 2 Unscrew injection line at cold start nozzle.



- 3 Detach engine hood and remove intake hose.

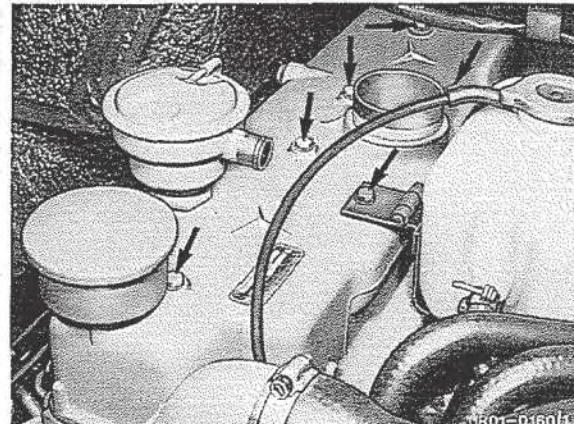
Note: For engine model 353.959, detach connection for manifold pressure compensator at cylinder head cover, intake hose at engine breather filter as well as the connection to the turbocharger.



- 4 Unscrew and remove cylinder head cover.
- 5 Clean sealing surface on cylinder head and cylinder head cover.
- 6 Install in reverse order, using new gasket.
Tightening torques, see 1.4/1.

Note: The fastening bolts of the cylinder head cover must be fully screwed in so that the head of the bolt makes contact with the cover while ensuring that the bracket for the coolant equalization tank is secured with a long bolt.

- 7 Start engine and check for leaks.



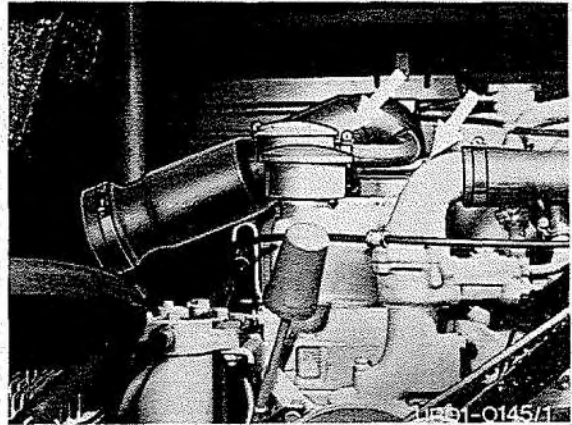
Removal

Note: Make sure that engine is cold.

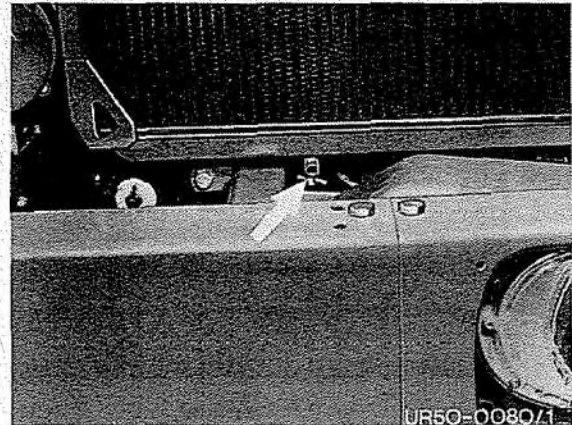
Removal is similar for both engine model No. 353.959 and No. 353.961.

1 Prop up cab and clean upper parts of engine. See 60.5-2.1/1, or 60.5-2.2/1.

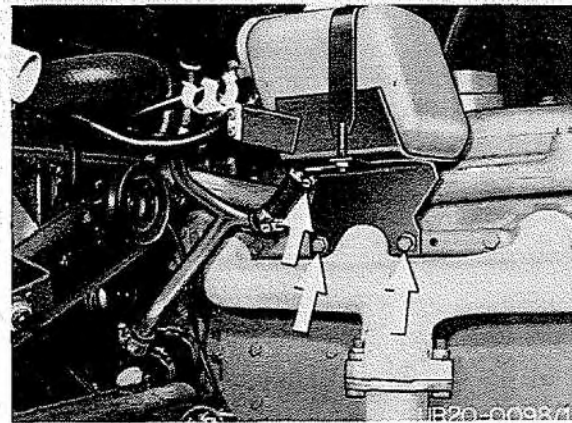
2 Detach intake pipe.



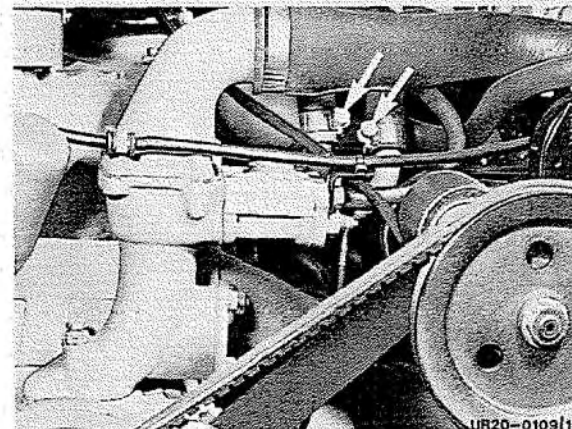
3 Drain coolant.

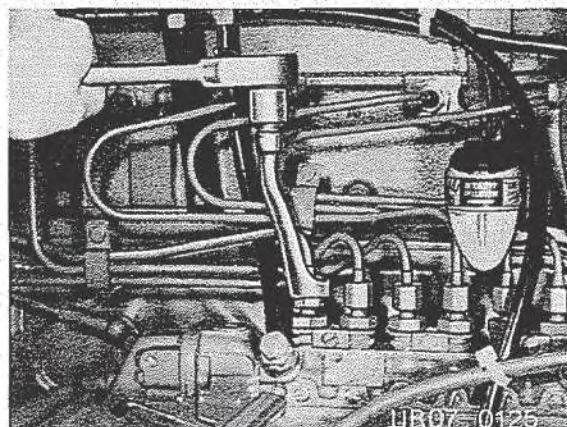


4 Detach coolant reservoir and bracket.

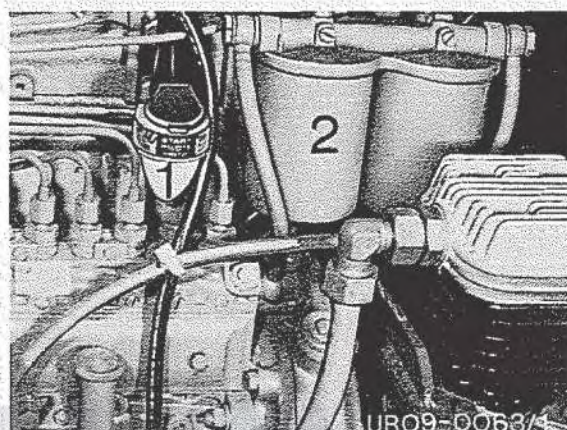


5 Detach heater hoses.

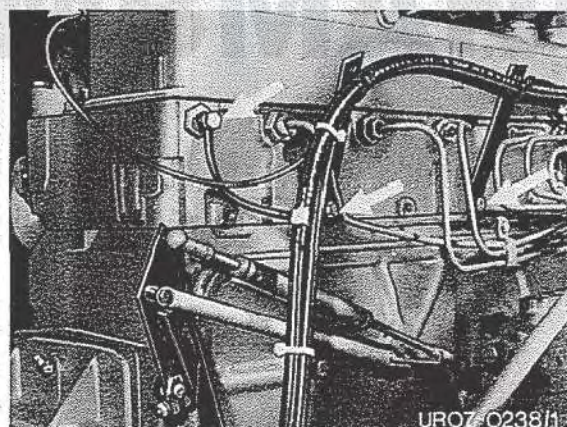




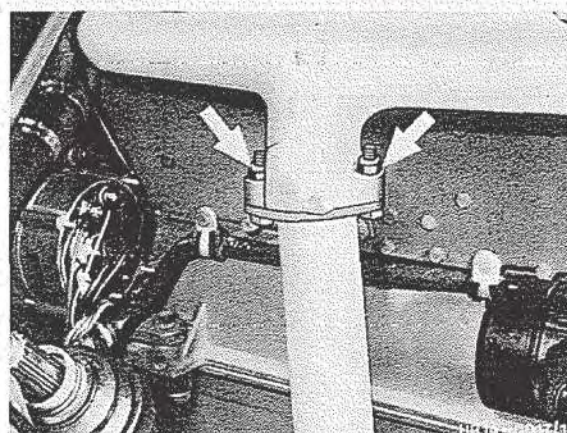
- 6 Using special tool No.14 , detach injection lines at injection pump.



- 7 Detach cold starting reservoir (1) and double fuel filter (2).

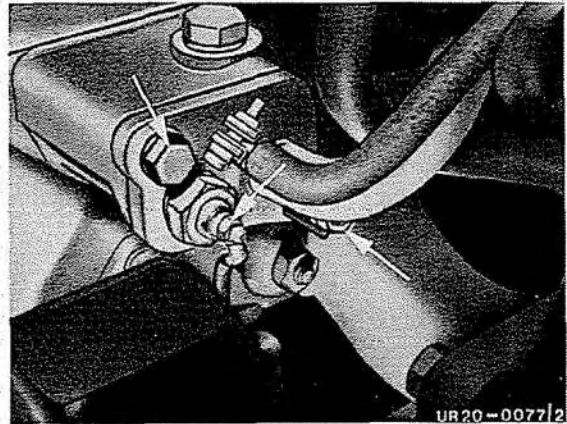


- 8 Detach cold starting line, leak-off line and bracket for fuel lines at cylinder head. Tie up at one side.

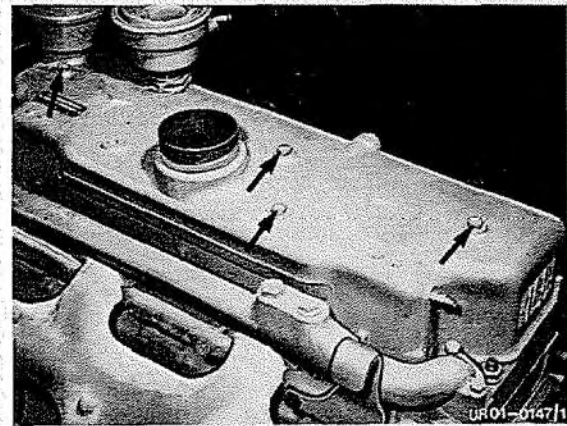


- 9 Detach exhaust pipe from exhaust manifold.

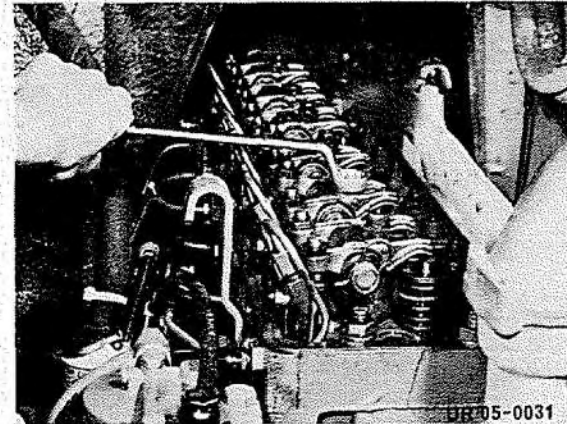
- 10 Disconnect coolant temperature pickup and detach coolant pipe at coolant thermostat body.



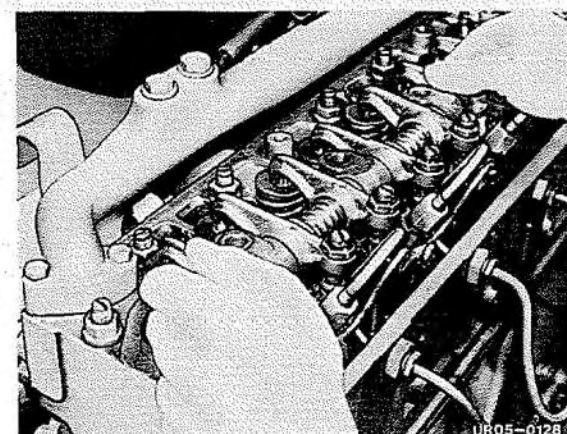
- 11 Remove cylinder head cover.

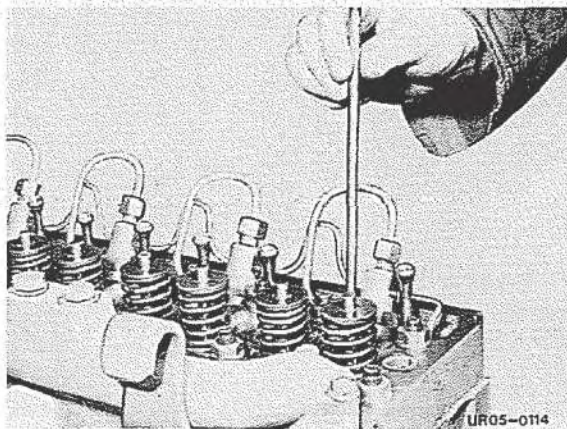


- 12 Remove bolts holding rocker arm brackets.



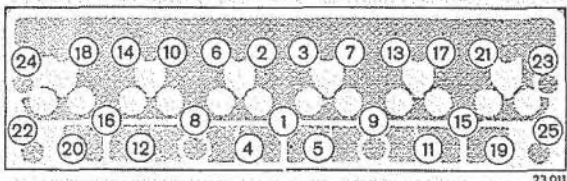
- 13 Remove rocker arm brackets with rocker shafts, connecting nipples and rocker arms.



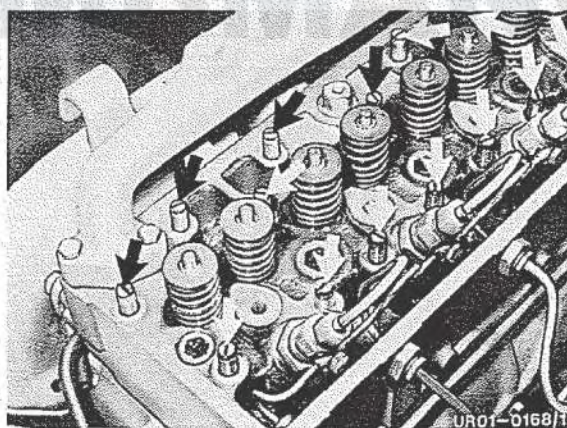


14 Carefully withdraw pushrods.

Note: To avoid withdrawing the tappets from their guides, the tappets must previously be loosened by a short, sharp up and down movement of the pushrods.

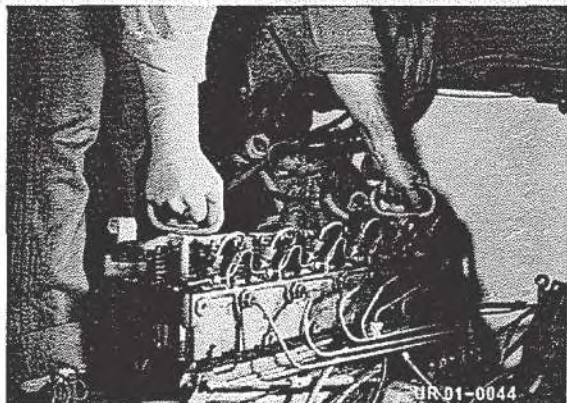


15 Undo all the cylinder head nuts or cylinder head bolts in the reverse order to the tightening chart.



16 Remove studs.

Note: The studs and cylinder head nuts fitted previously were replaced as from engine end No. 678 155 by cylinder head bolts which can be exchanged in reverse.



17 Using special tool No. 18, lift up cylinder head and remove gasket.

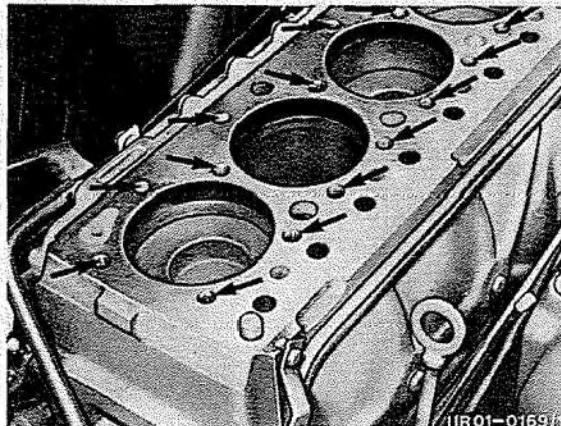
Important!

Deposit cylinder head so that no damage is done to injection nozzles.

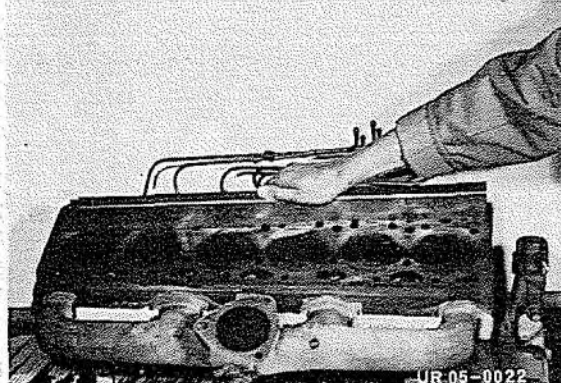
18 Clean cylinder head and crankcase parting surfaces

Important!

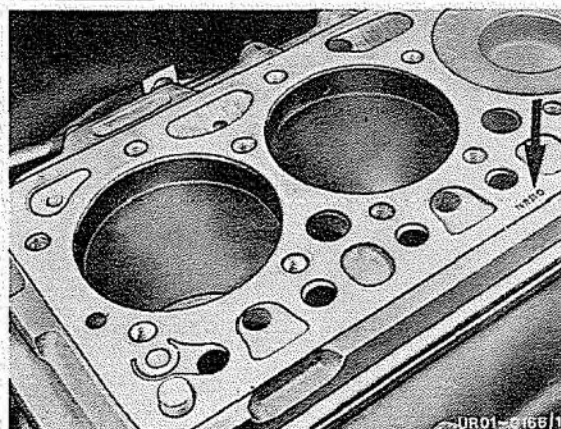
Blind holes in the crankcase **must not** show any signs of dirt or oil.

19 Check parting surfaces with straight edge.
See table on page 1.2/3 for permissible unevenness.

Note: Distorted or damaged parting surfaces should be refaced to the necessary peak-to-valley height.
See table on page 1.2/2 and 1.2/3 for permissible peak-to-valley height.

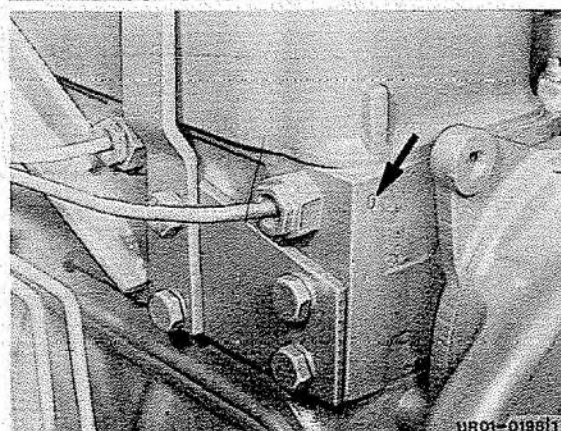
**Installation**

1 Position new cylinder gasket on crankcase with lettering facing upwards



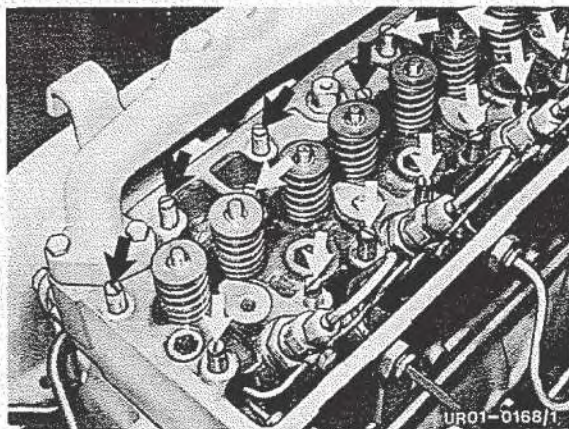
2 Using special tool No. 18, position cylinder head on crankcase.

Note: For engine model No. 353.959, a modified cylinder head marked with "O" has been installed from engine end No. 691 150. The modified cylinder head **cannot** be retrofitted on previous models.



01.8 Removal and Installation of Cylinder Head

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To Engine End No. 678 154

3 Oil threads of stud bolts and screw in.

Important! Do not pour oil into the threaded holes.

4 Fit cylinder head nuts and tighten referring to tightening diagram.

1st tightening stage 60 Nm

2nd tightening stage 90 Nm

3rd tightening stage 110 Nm

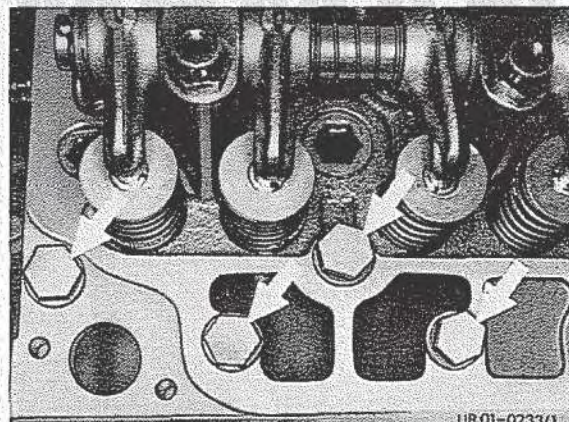
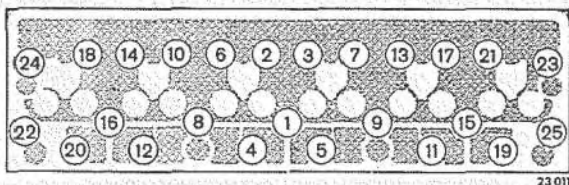
Note: Tighten cylinder head nuts for the first time to 120 Nm at the 3rd tightening stage.

From Engine End No. 678 155

5 Oil cylinder head bolts at the thread and at the head contact surface and screw in,

Important! Do not pour oil into the threaded holes.

Note: Only cylinder head bolts are stocked in the spare parts store. The cylinder head bolts can be installed in engines with a lower engine end number.



6 Tighten cylinder head bolts according to tightening diagram.

1st tightening stage 60 Nm

2nd tightening stage 90 Nm

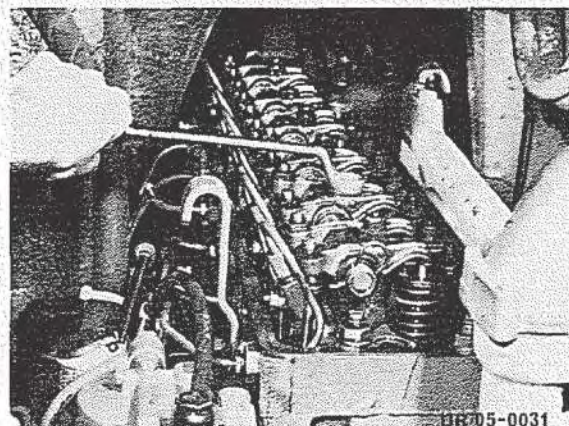
3rd tightening stage 110 Nm

Note: Tighten cylinder head bolts for the first time to 120 Nm at the 3rd tightening stage.

7 Install push rods.

8 Attach rocker arm brackets together with rocker arm shafts and rocker arms. Turn back valve adjusting screws and tighten rocker arm brackets.

Tightening torques see 1.4/1.

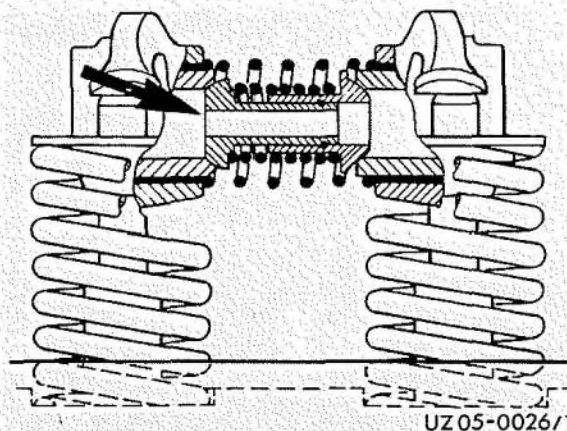


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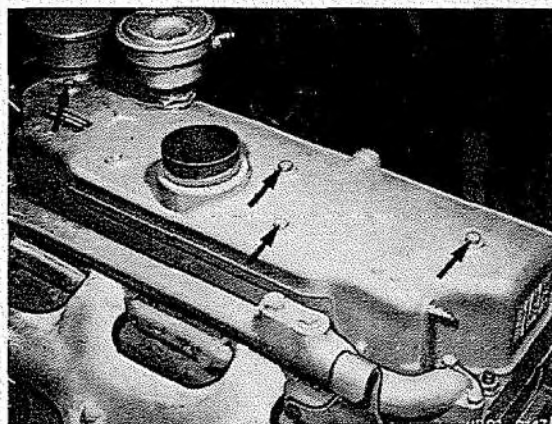
9 Using special tool No. 2, install connecting nipple and spring.

Note: Install the connecting nipple with the **small** inside diameter **in the direction of driving**. Ensure it is correctly seated.

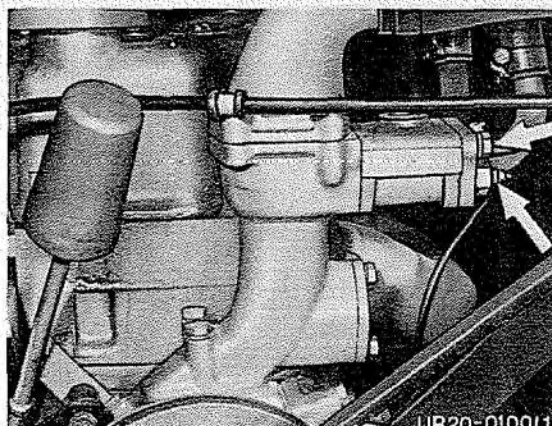
10 Adjust valve clearance, refer to 05.8 - 2.1/1.



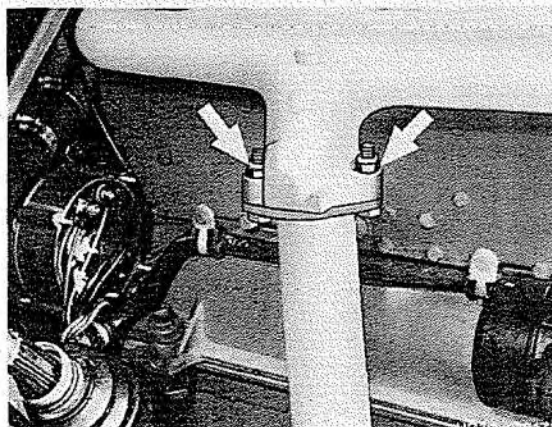
11 Install cylinder head cover.
For tightening torques, see 1.4/1.



12 Attach coolant pipe, with new gasket on coolant regulator housing, connect coolant temperature sensor cable.
For tightening torques, see 1.4/1.

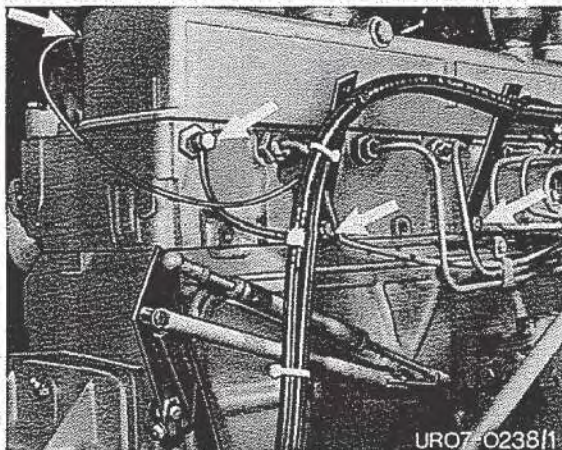


13 Attach exhaust pipe with gasket to exhaust manifold.
For tightening torques, see 1.4/1.

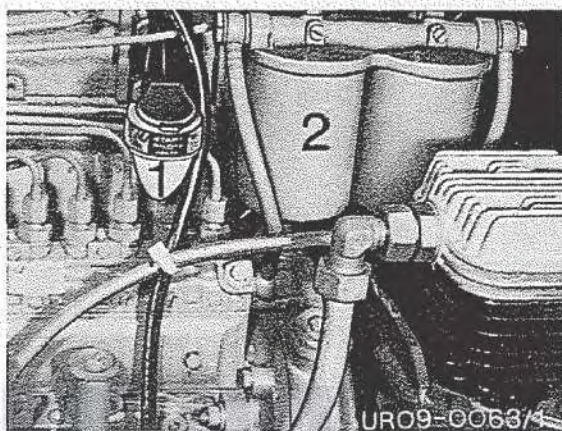


01.8 Removal and Installation of Cylinder Head

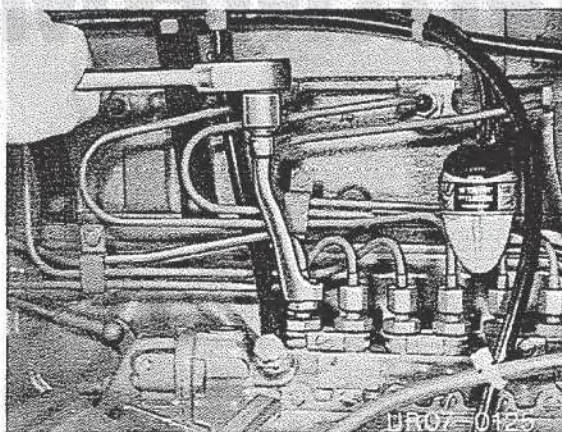
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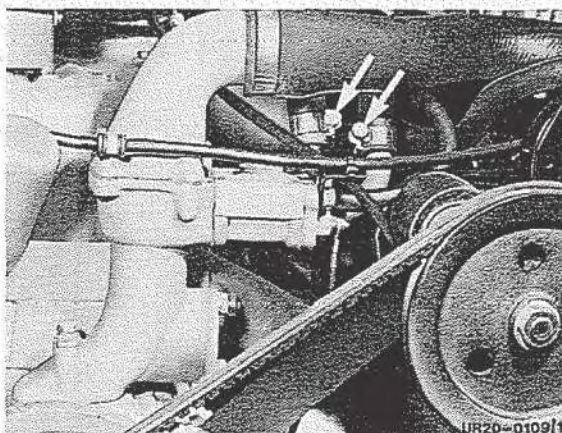
14 Attach cold starting line, leakage fuel line and brackets for fuel lines to cylinder head.



15 Install cold start reservoir (1) and double fuel filter (2).

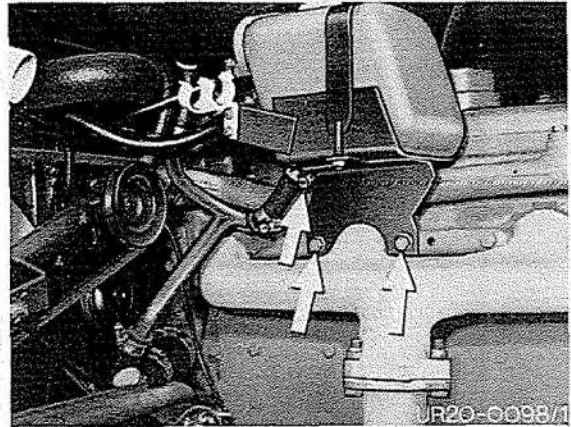


16 Using special tool No. 1, connect injection lines to injection pump and tighten with special tool No. 15. For tightening torques, see 1.4/1.



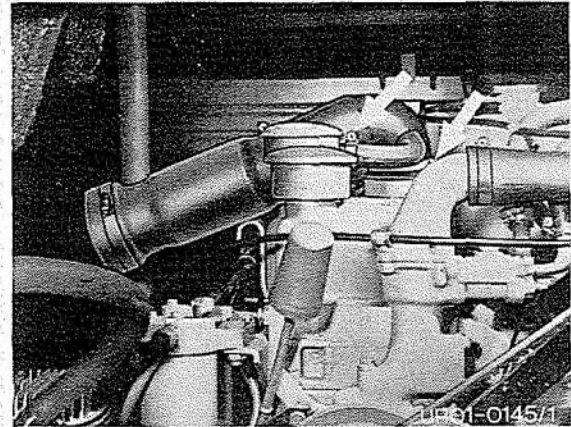
17 Attach heater hoses.

18 Attach coolant equalization tank with bracket.



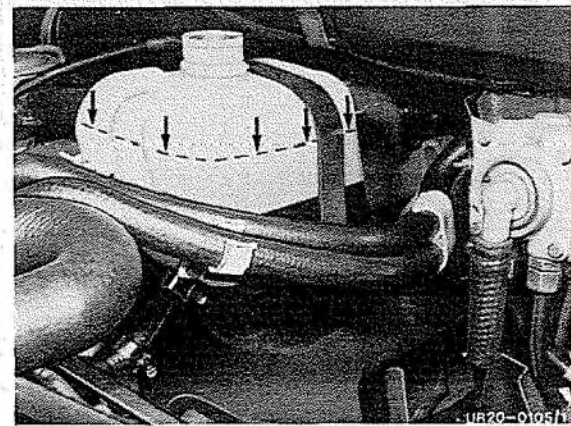
19 Attach intake pipe.

20 Lower driver's cab, see 60.5-2.1/1 or 60.5-2.2/1.



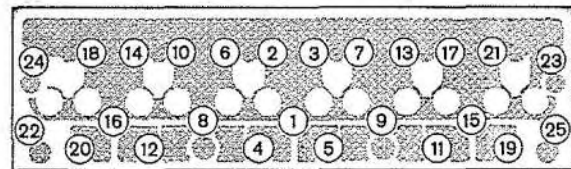
21 Fill with coolant, check cooling system for leaks. For capacities see 1.3/2.

Note: The equalization tank should be two-thirds full.



Important!

Upon completion of the installation, warm up engine, loosen each nut individually in sequence shown in the tightening diagram and then finally tighten to a torque of 110 Nm. On no account must all nuts be loosened simultaneously before final torque is applied. Readjust valve clearance. After 500 km, retighten the cylinder head nuts and readjust the valves.



23 011

Disassembly

1 Remove cylinder head, see 4.1/1.

2 Lay cylinder head on a support.

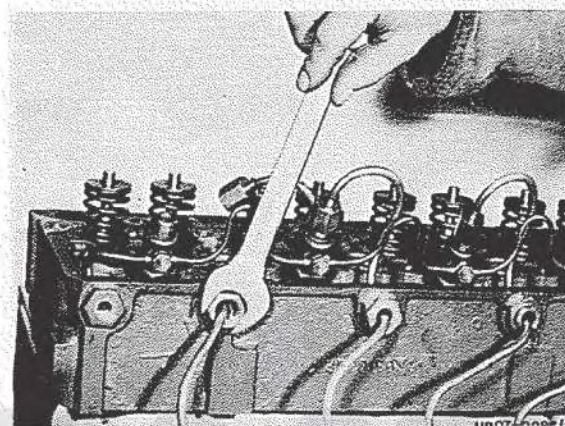
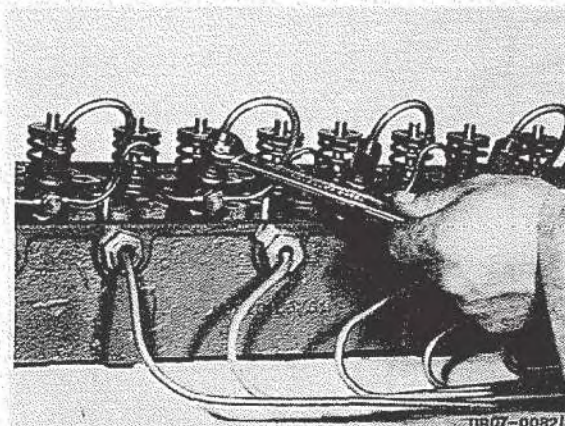
Important!

The cylinder head must not rest on the injector nozzles.

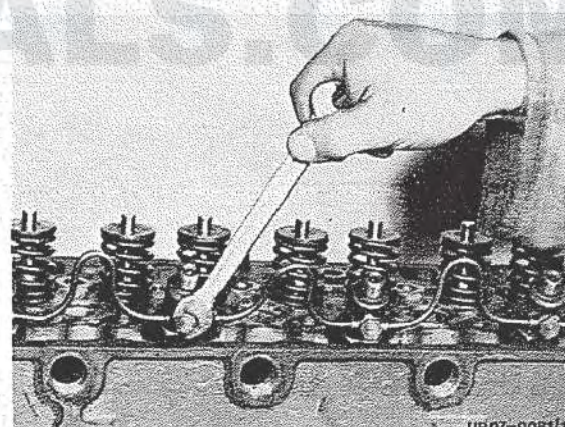
3 Remove exhaust manifold and coolant rail from cylinder head.

4 Remove injector pipes from injectors with special tool No. 1.

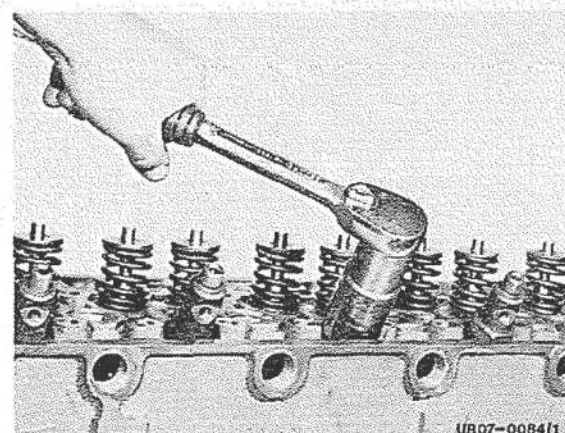
5 Release injector pipes from cylinder head and remove. Do not bend injector pipes.



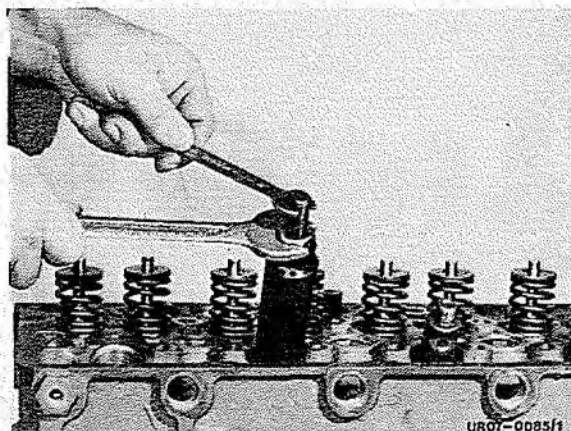
6 Remove leak-off pipes.



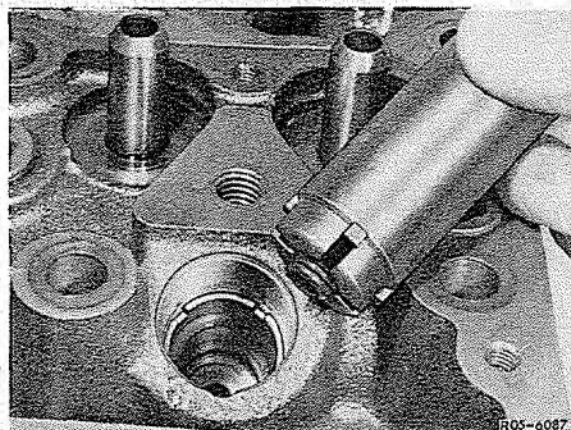
7 Remove thrust screw for nozzle holder with special tool No. 4.



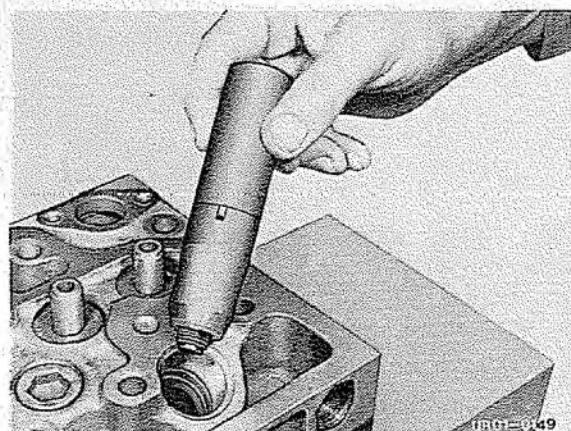
352/352 A



- 8 Remove nozzle holder with special tool No. 5.
- 9 Remove valves, see 05.8-2.2/1.

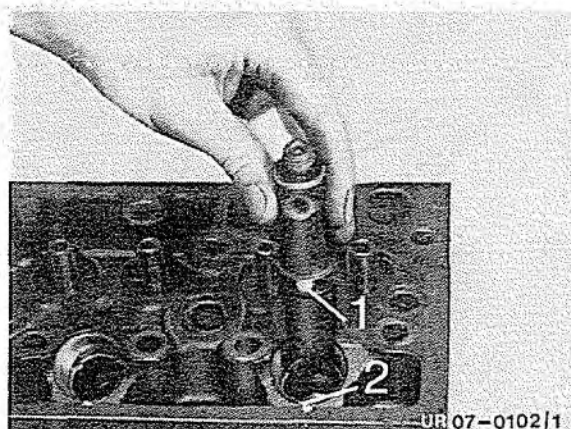


- 10 Remove protective sleeve with special tool No. 3.
- 11 Clean all parts, check and change if necessary.
- 12 Check valve guides and change if necessary, see 6.1/1.
- 13 Check valve seats and reface or renew if necessary. See 6.2/1 or 6.3/1.
- 14 Check injector nozzles, see 07.8-4.2/1.



Assembly

- 1 Insert new sealing ring for the protective sleeve into cylinder head.
 - 2 Insert protective sleeve and tighten with special tool No. 3.
- Tightening torque, see 1.4/1.

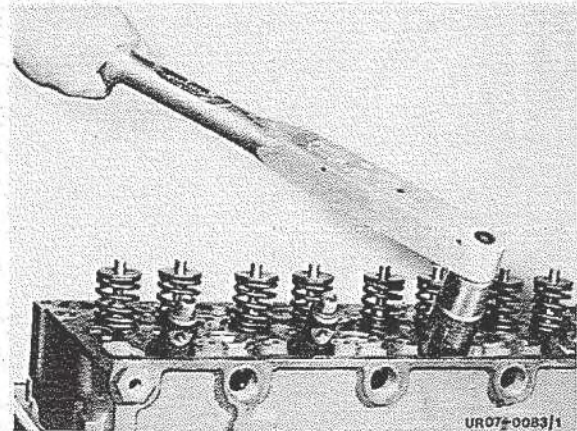


- 3 Insert new sealing ring into the protective sleeve
- 4 Fit nozzle holder with nozzle into cylinder head observing lock on the nozzle holder (1) and groove (2) in the cylinder head.

Important!

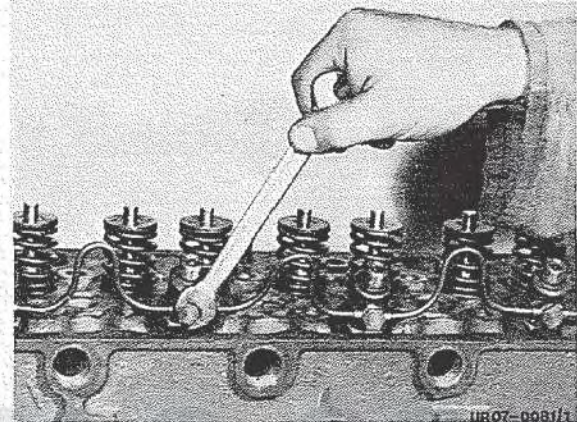
Do not knock nozzle.

- 5 Fit valves, see 05.8-2.2/1.
- 6 Tighten thrust screw for injector nozzle with special tool No. 4.
Tightening torque, see 1.4/1.

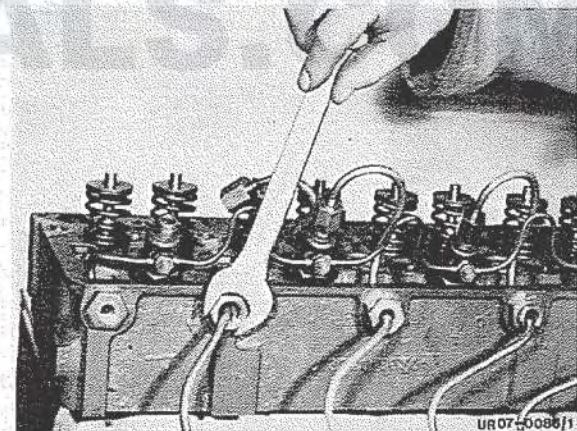


- 7 Fit leak-off pipes with new sealing pieces.
Tightening torque, see 1.4/1.

Note: Fit locking bolt (24 mm) with sealant No. 5.



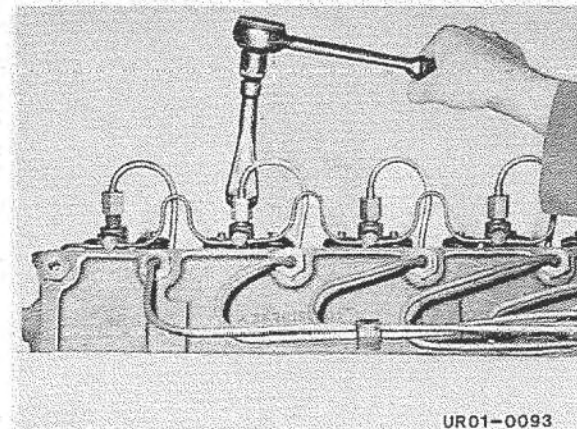
- 8 Fit injector pipes, fit locking bolt (24 mm) with sealant No. 5, renew rubber seals at the bushings and tighten locking bolt (17 mm).



- 9 Tighten without tensioning the injector pipes using special tool No. 15.
Tightening torque, see 1.4/1.

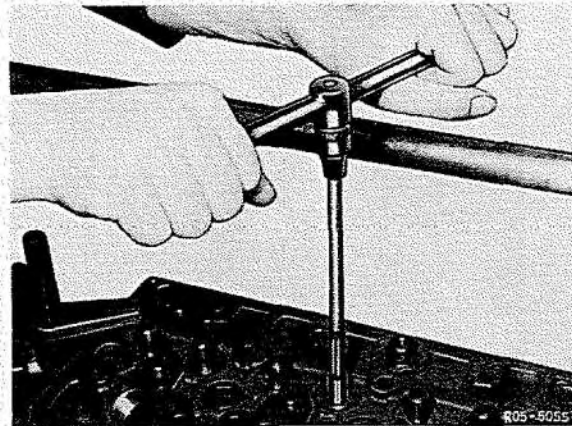
- 10 Fit exhaust manifold and coolant rail with new gaskets.
Tightening torque, see 1.4/1.

- 11 Install cylinder head, referring to 4.1/1.

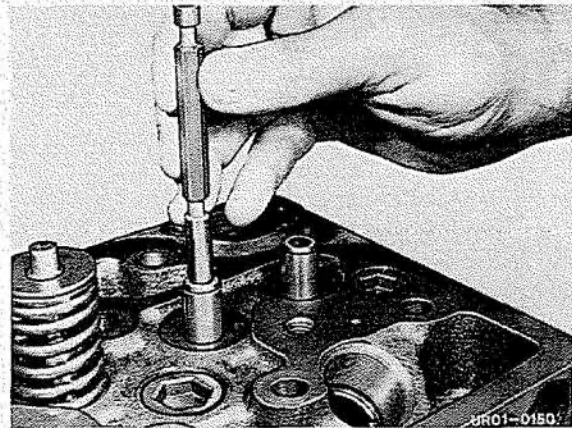


Checking

- 1 Disassemble cylinder head, referring to 5.1/1.
- 2 Clean valve guide, removing carbon deposits with special tool No. 11 or No. 12,

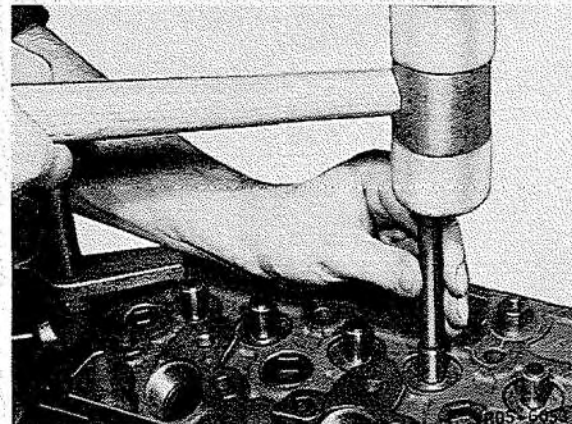


- 3 Check bore using special tool No. 16 or No. 18.

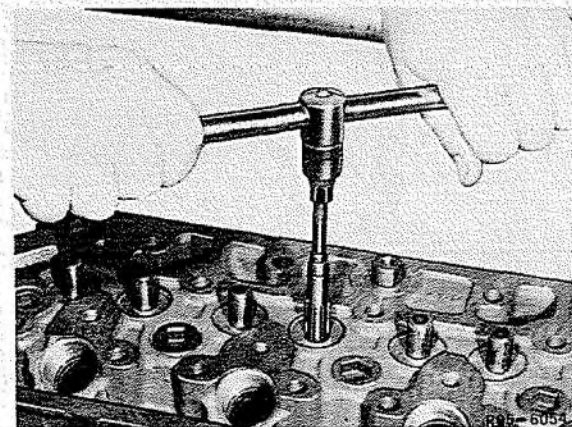


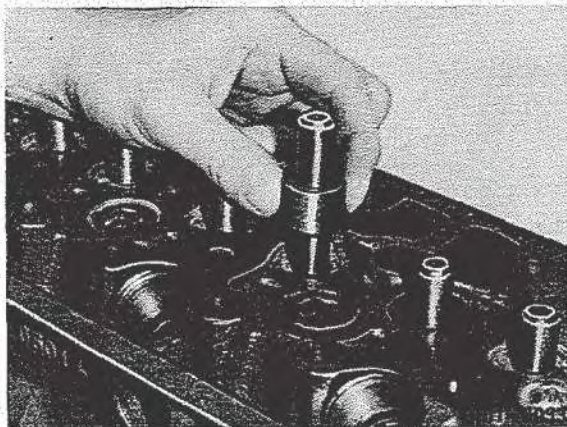
Exchanging

- 1 Drive out valve guide using special tool No. 7 or No. 8.
- 2 Clean and check basic bore in cylinder head. See 1.2/4 for table.



- 3 If "repair stage" valve guides are to be fitted, ream basic bore to next higher repair stage, using special tool No.13 . See 1.2/4 for repair stages.





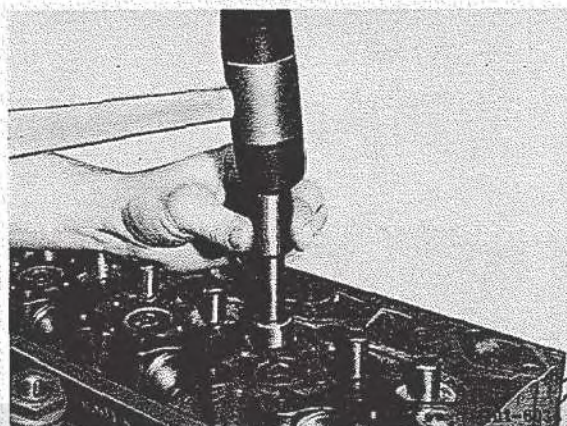
4 Select valve guide so that overlap specification is met.

See table on page 1.2/4.

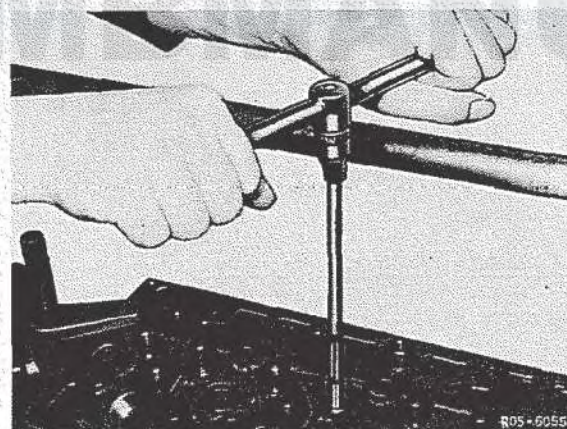
5 Heat cylinder head.

6 Coat new valve guide with graphite oil and position in cylinder head bore.

7 Slip spacing sleeve of special tool No. 10 over valve guide.

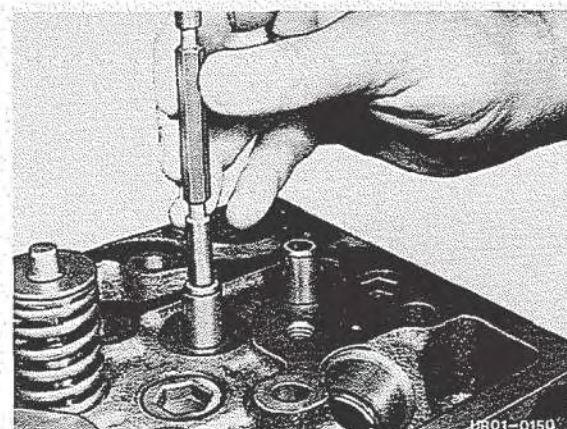


8 Knock valve guide into position, using special tool No. 10.



9 Check valve guide for firm fit, either by knocking with hammer or by applying test pressure. Loose valve guides are to be knocked out again and replaced by valve guides of larger outside diameter.

10 Ream inside diameter of valve guide, using special tool No. 11 or No. 12.



11 Check inside diameter of valve guide, using special tool No. 16 or No. 18.

12 Re-assemble cylinder head, referring to 5.1/1.

Refacing

- 1 Disassemble cylinder head, referring to 5.1/1.
- 2 Check valve guides and exchange if necessary, referring to 6.1/1.
- 3 Clamp cylinder head in jig.
- 4 Check valve seat.

Note: If valve seats show only slight traces of wear, they can be refaced without necessitating replacement of valve seat rings, provided they are not machined below their bottom limit.
See 1.2/4 for repair size.

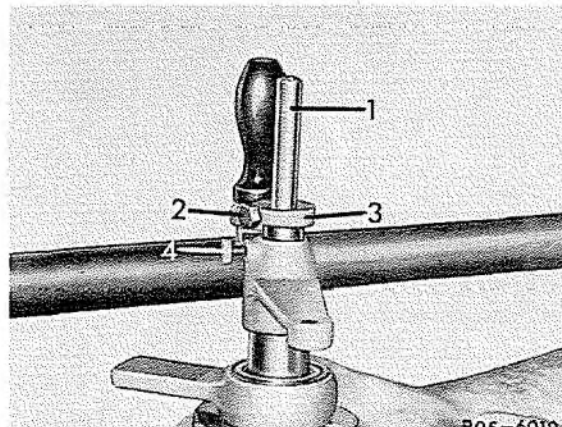
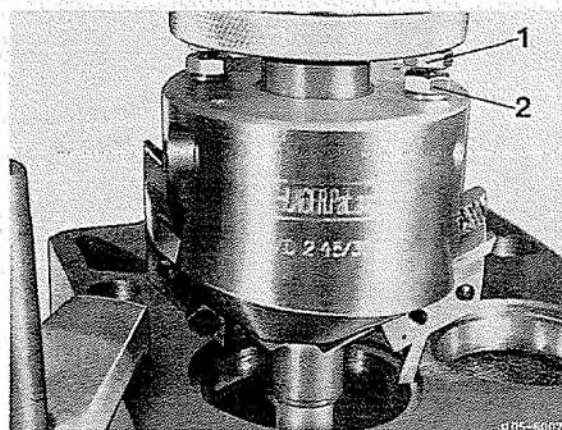
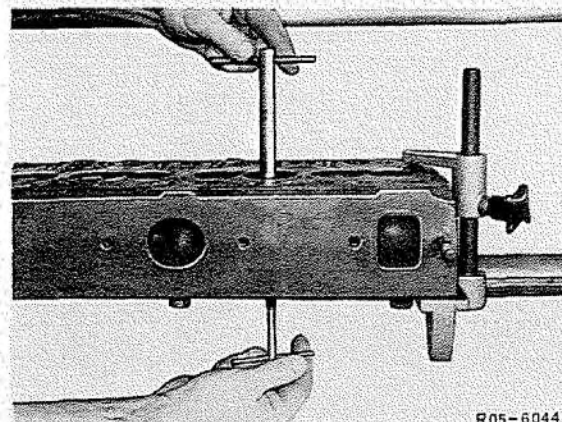
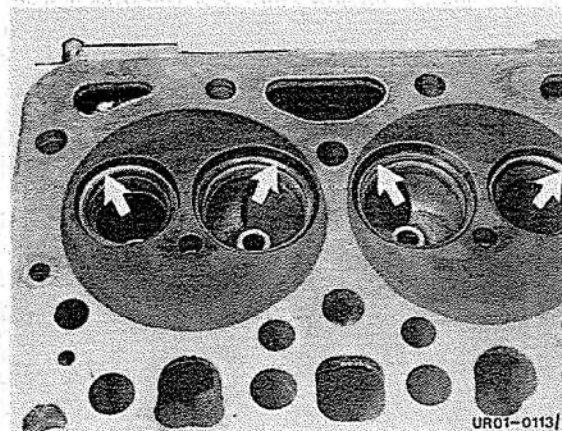
- 5 Introduce pilot into valve guide so that stop of slotted rod rests on valve guide. If necessary, press slotted rod downwards using screwdriver, and tighten pilot.

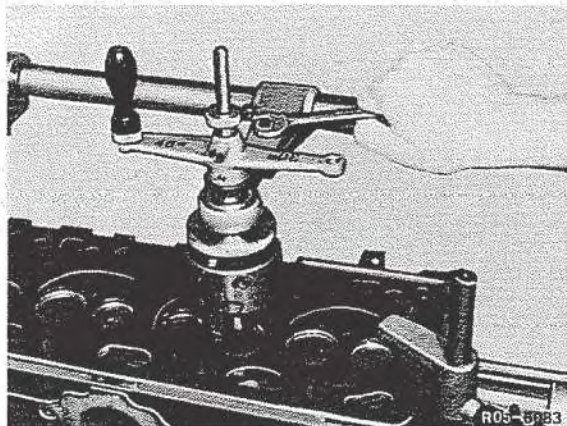
- 6 Fasten cutting tool to slide rest.

Note: Make sure that cutting tool is positioned correctly on slide rest. Release coupling nut to stop free slide rest. Attach crank handle to arm marked 45°.

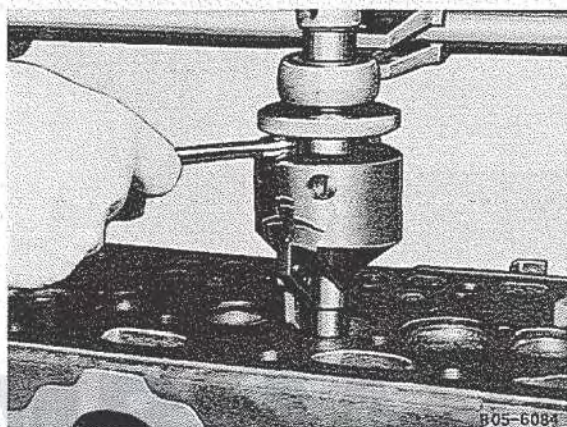
- 7 Release coupling nut (2), slip cutter over pilot and turn screw (1) until cutting tool is positioned in center of valve seat.

- 8 Press pilot support (1) downward onto pilot, and clamp with screw (2), making sure that knurled ring (3) is screwed down and locking screw (4) is tight

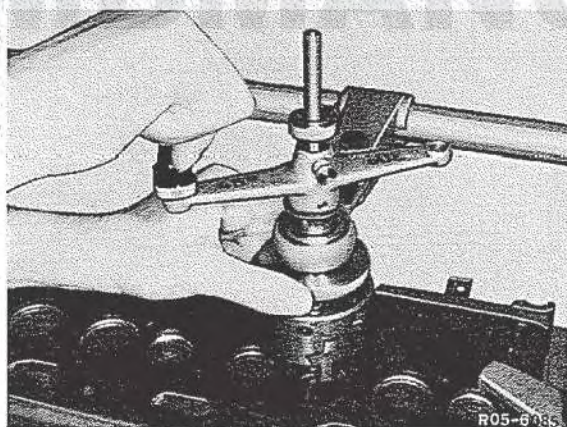




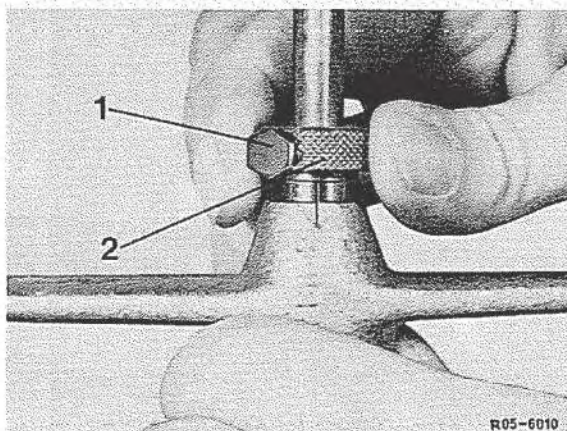
9 Turning hand crank, move steady bar to correct position. Floating guide is to be locked horizontally in roughly the center of the guide with the aid of steady clamp. Cutter must turn just as smoothly as before.



10 Turning the quick-action adjuster sets cutting tool to inner seat edge. Tighten coupling nut now, but do not start chip feed yet.



11 Turn crank while holding feed screw. Cutting is normally irregular at this position. After turning once release coupling nut of quick-action adjuster and move tool back to inner setting.



12 Release locking screw (1) and turn knurled collar (2) anti-clockwise by about 1/2 to 2 divisions (1 division = 0.1 mm). Retighten locking screw (1) and coupling nut, then beginning to turn again.

13 Hold feed screw and turn crank until cutting tip has reached outer position.

14 Chip feed is to be continued until seat is fully refaced (do not go below machining limit). Then turn once again without chip feed. Newly fitted valve seat rings must show values given in table. See 1.2/4 for repair sizes.

15 Re-assemble cylinder head, referring to 5.1/1.

Refacing

1 Disassemble cylinder head, referring to 5.1/1.

2 Check valve guides and exchange if necessary, referring to 6.1/1.

3 Check valve seat.

Note: If valve seats show only slight traces of wear, they can be refaced without necessitating replacement of valve seat rings, provided they are not machined below their bottom limit.
See 1.2/4 for repair size.

4 Introduce special tool No. 14 into valve guide.

5 Position special tool No. 14 and adjust cutter to seat diameter.

See table on page 1.2/3.

Note: For valve seats with 45° seat angles the 45° side of the cutter head is for cutting the seat whereas the 30° side is for the clearing cut. For valve seats with 30° angles, the two sides are to be reversed.

6 Reface valve seat, applying gentle pressure.

7 Measure valve seat runout.

See table on page 1.2/3.

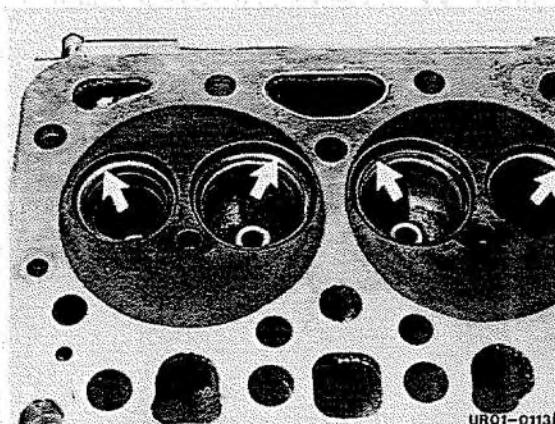
Important:

Check directly after finishing the valve seat. The guide drift must not be released or withdrawn beforehand.

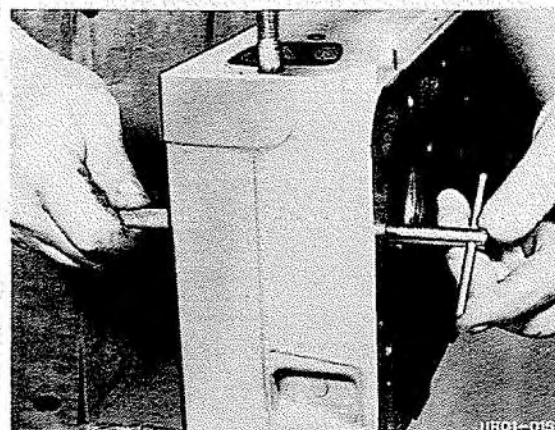
8 Measure valve seat width; if necessary, correct to 70° at top and bottom.

See table on page 1.2/3.

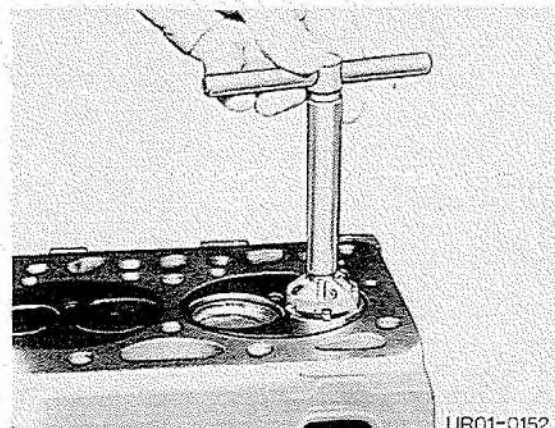
9 Re-assembly cylinder head, referring to 5.1/1.



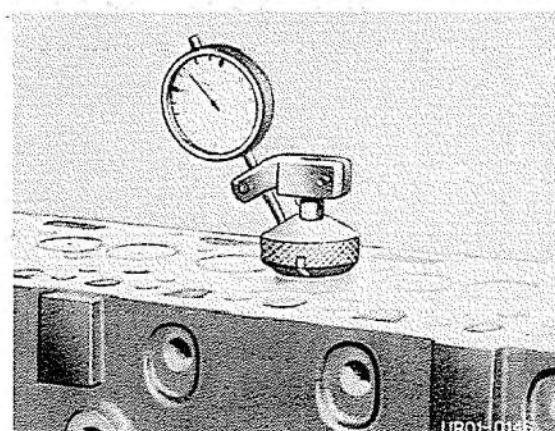
UR01-0113/1



UR01-0151



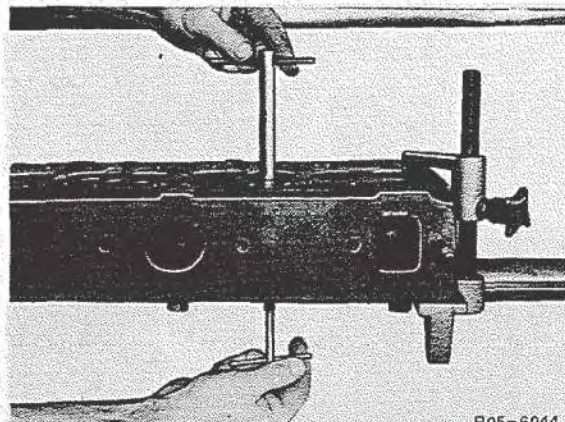
UR01-0152



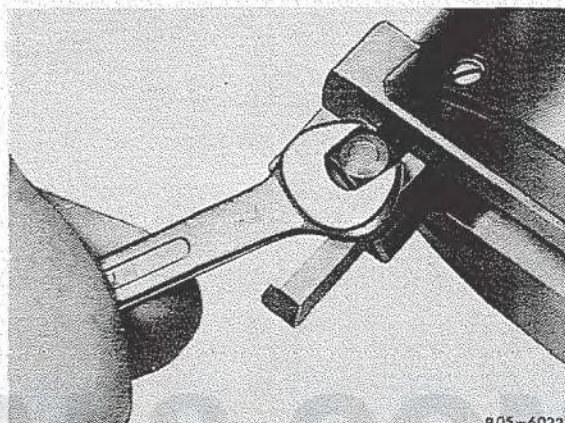
UR01-0145

Removal

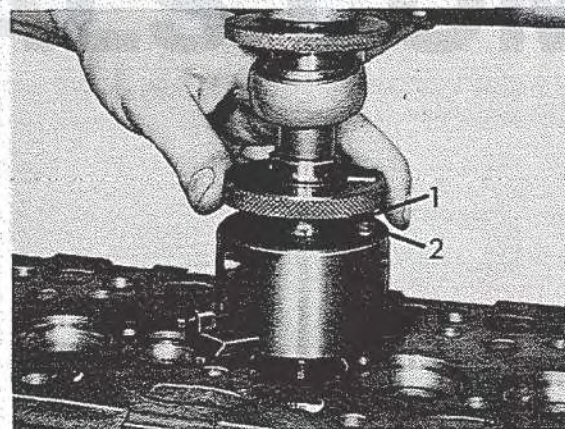
- 1 Disassemble cylinder head, referring to 5.1/1.
- 2 Check valve guides and renew if necessary, referring to 6.1/1.
- 3 Clamp cylinder head in jig.
- 4 Introduce pilot into valve guide until stop on slotted rod rests on valve guide. If necessary, force slotted rod downwards using screwdriver. Tighten with drift, inserting into pilot from above and below.



- 5 Clamp groove cutter in slide rest.

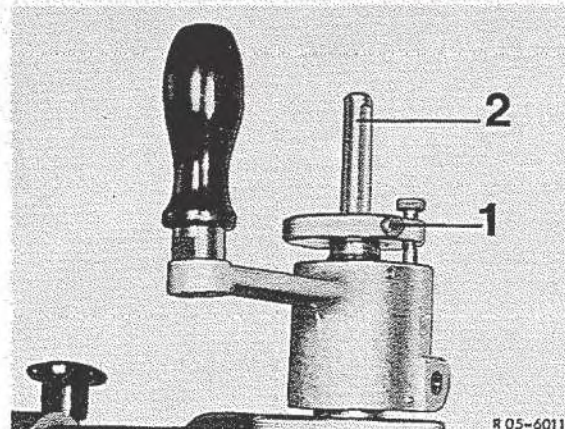


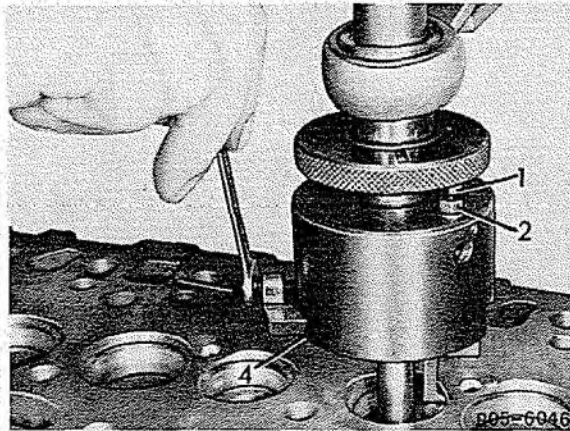
- 6 Release lock-nut (2). Moisten pilot with oil, slip cutter over pilot, turn screw (1, quick-action adjustment) until tool contacts pilot and then slide cutter downwards until tool is located in center of valve seat ring.



- 7 Hold cutter at this position. Release locking screw (1) of pilot support, and press pilot support (2) downwards until it contacts pilot. Retighten locking screw (1).

- 8 Turning hand crank, move steady bar to suitable position relative to work. Lock floating guide horizontally in roughly center of guide with aid of steady clamp. Cutter must now turn just as smoothly as before.

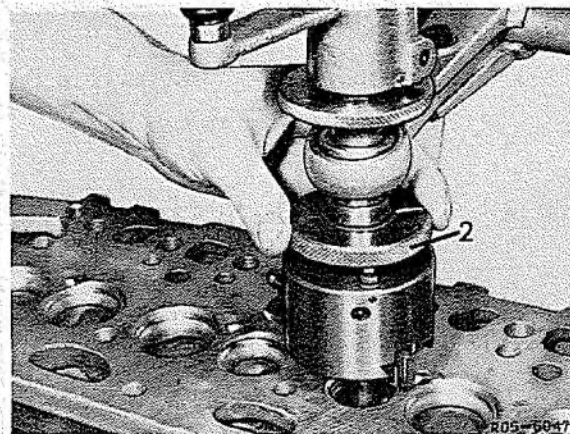




9 Slacken screw of quick-action adjuster (1) until tool has reached, yet does not yet touch valve seat ring. Tighten lock-nut (2). Turn horizontal stop screw (3) fully home against body and then slacken by 2 to 3 mm. Tighten clamping screw (4) beneath.

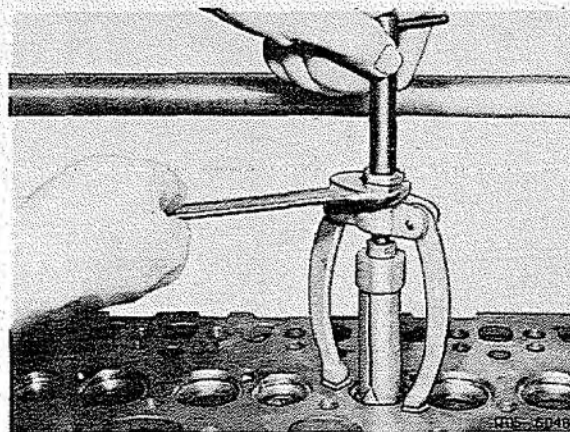
10 Adjust tool height so that about 1 mm metal is left on valve seat ring at bottom after machining.

Note: Hold knurled collar (2) and turn cutter crank clockwise, so moving tool downwards. Turning anti-clockwise serves to raise tool.



11 Turn tool and simultaneously hold knurled collar (2) which has to be briefly released as torsion increases slightly. This serves to turn groove in valve seat ring to depth of about 2 to 3 mm.

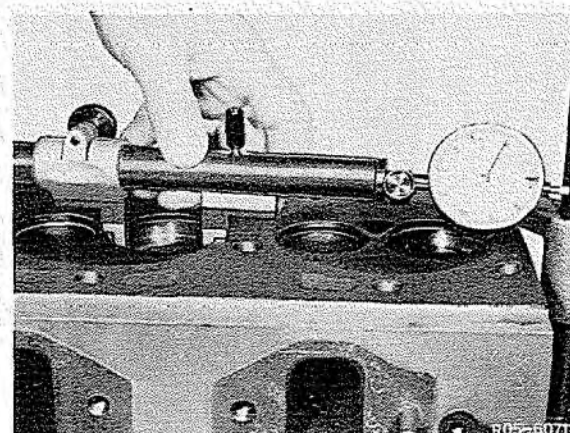
Note: Avoid coarse cuts. Cutter must always turn easily, this being achieved by briefly releasing knurled collar (2).



12 Remove cutter.

13 Fit internal remover in annular groove, tighten nut and draw out using counter-support.

Note: Place copper strips beneath counter-support to avoid damaging cylinder head face.



14 Adjust internal measuring instrument using micrometer.

15 Measure bores for valve seat rings in cylinder head, using internal measuring instrument.

Note: Holes are to be bored to next step should table values not be reached.
See 1.2/4 for repair stages.

16 Remachine basic bore, referring to 6.5/1.

Installation

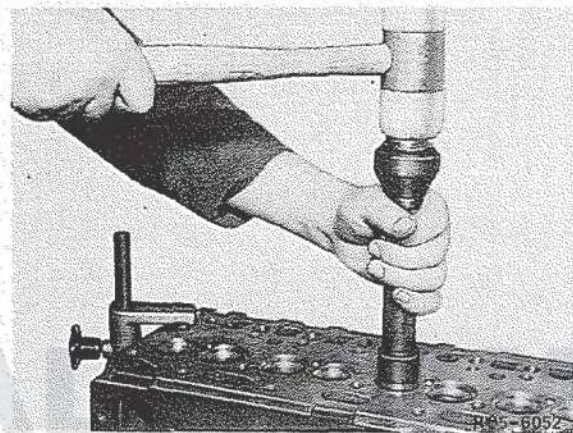
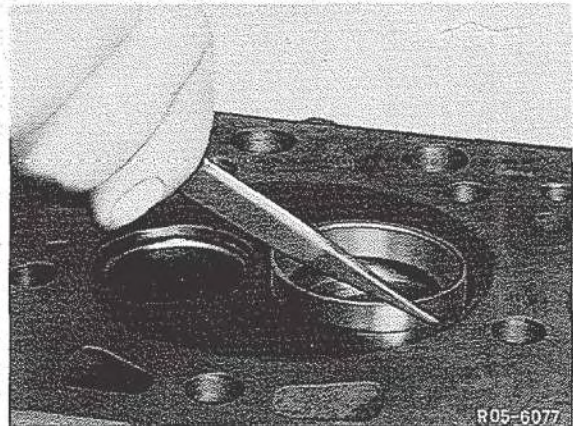
1 Place valve seat rings in container full of liquid oxygen, allowing rings to shrink for about 20 to 30 minutes.

2 Heat cylinder head to 80°C.

3 Take valve seat rings out of liquid oxygen and place on bore of heated cylinder head.

4 Knock valve seat rings into position, using special tool No. 9.

Note: Do not delay in fitting the valve seat rings. Do not touch liquid oxygen or come into contact with freezing valve seat ring.



MBMANUAL

Remachining basic bore

- 1 Clamp tool in slide rest.
- 2 Introduce pilot into valve guide until stop on slotted rod rests on valve guide. If necessary, force slotted rod downwards using screwdriver. Tighten with drift, introducing into pilot from above and below.

- 3 Release lock-nut (1), moisten pilot with oil, slip cutter over pilot, turn quick-action adjusting screw (2) until tool moves horizontally beyond bore, and then slide cutter downwards until tool rests on cylinder head.

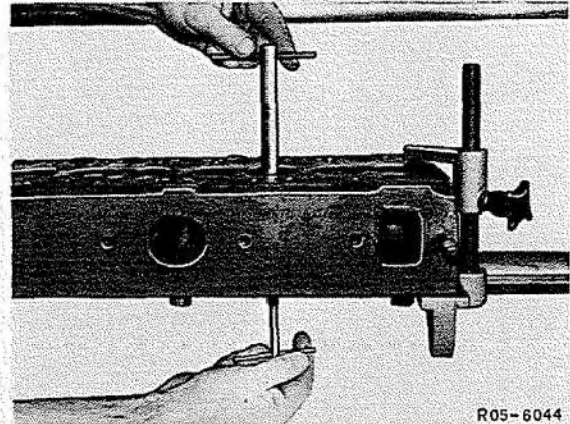
Important:

Fit tools carefully so as not to damage carbide tip.

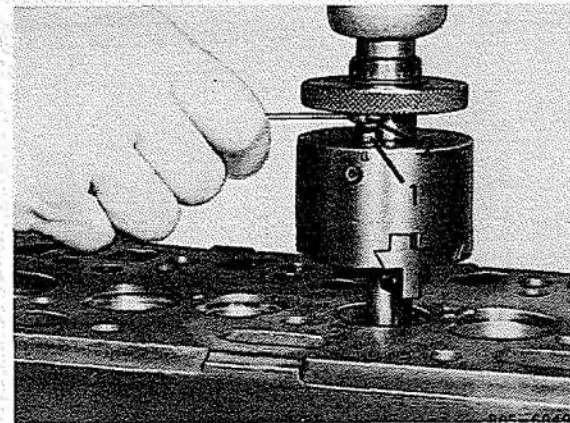
- 4 Release locking screw (1), force pilot support (2) downwards until it contacts pilot, and then re-tighten locking screw (1).

- 5 Turning the knurled collar serves to adjust tool height to point where it is just clear.

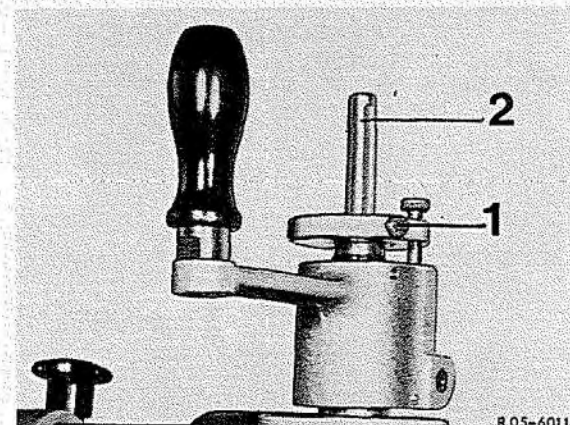
- 6 Lock floating guide horizontally with aid of steady clamp, making sure to set working depth between adjusting ring and floating guide. See 1.2/4 for working depth.



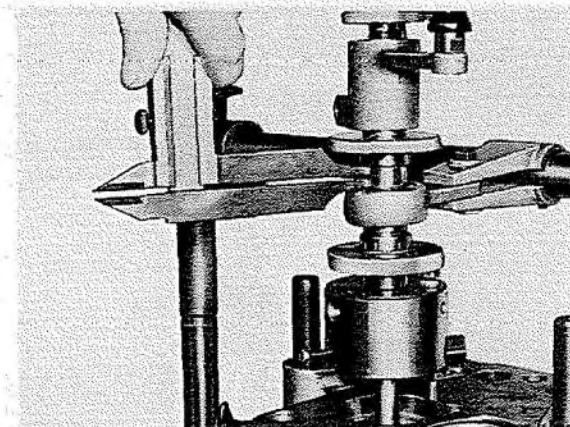
R05-6044

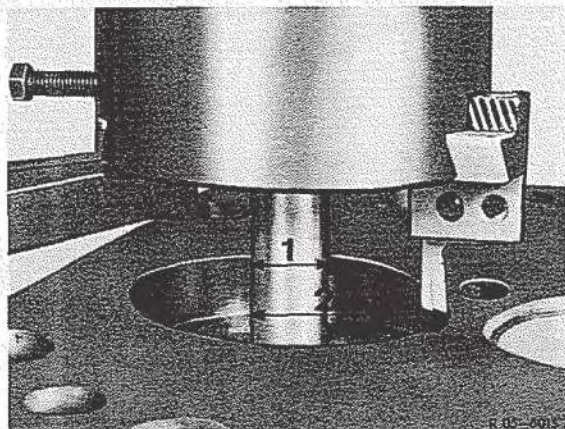


R05-6044



R 05-6011

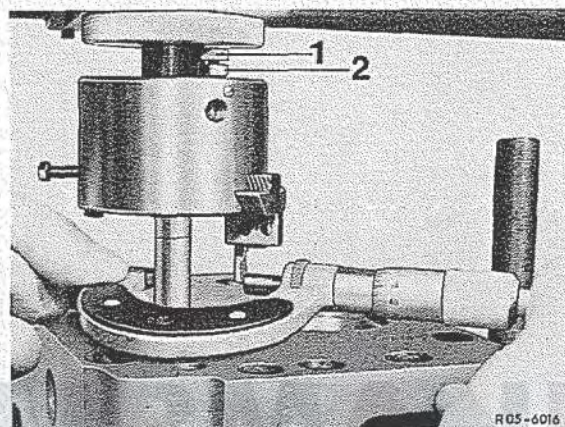




7 Calculate adjustment "2":

$$\text{"2"} = \frac{\text{Basic bore "c" or "d" + pilot dia. "1"}}{2}$$

See table 1.2/4 for "c" and "d"

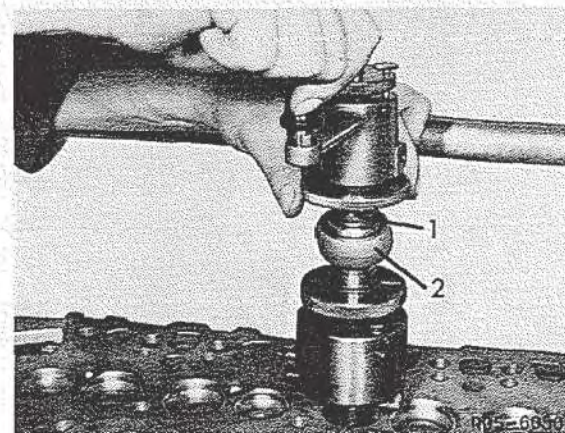


8 Release coupling nut, set micrometer to "2", slightly lift cutter, attach micrometer to pilot, and exactly adjust tool to size "2" using quick-action adjuster (1). Tighten coupling nut (2).

Note: You are advised to reduce tool setting by 0.1 mm in diameter for first cut.



9 Turn horizontal stop screw (1) fully home against body and tighten screw (2) beneath, so preventing stop screw from turning.



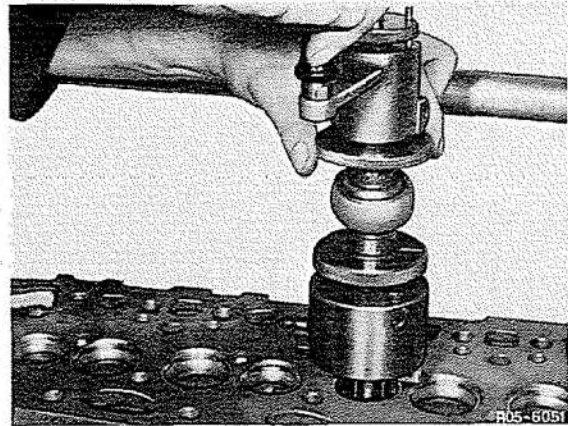
10 Turn hand crank and simultaneously hold upper knurled collar for vertical feed. Cut bore for valve seat ring until adjusting ring (1) contacts steady bearing (2).

Note: Keep adjusting tool until adjustment calculated in job No. 7 is reached. Then turn once without chip feed to obtain peak-to-valley height of max. 0.006 mm.

11 Release coupling nut and draw back tool at quick-action screw, while slightly raising cutter.

12 Turn hand crank and simultaneously hold lower knurled collar for horizontal feed. Reface bottom end surface until stop screw contacts body.

13 Remove cutter, and measure bore (be sure to obtain valve seat ring/bore overlap).
See 1.2/4 for bore dimensions.



Checking

Note: Engine temperature is to be +50°C to 80°C.

1 Remove injection nozzle without protective bushing. See 07.8—4.1/1.

2 Introduce special tool No. 17 into protective bushing and tighten with thrust screw from nozzle holder.

3 Attach compression pressure tester to special tool No. 17.

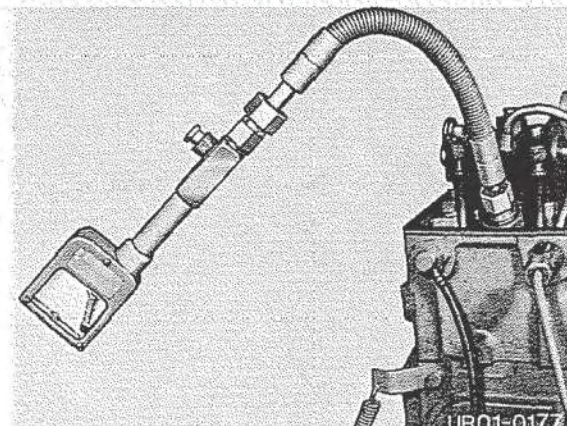
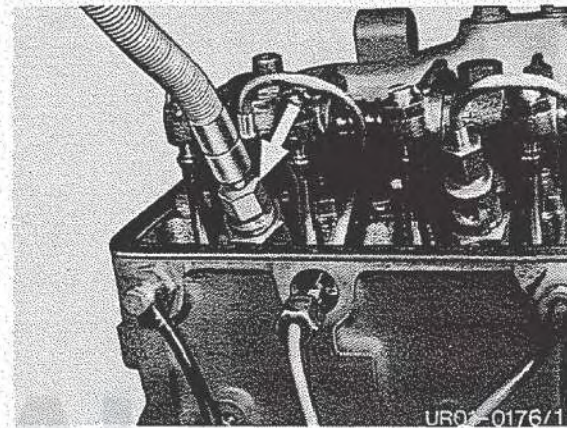
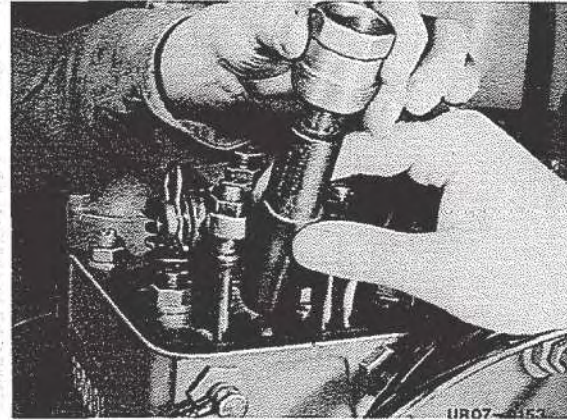
4 Place recording card in tester and set appropriate cylinder.

5 Move hand throttle to stop position and start engine several times.

6 Note pressure reading on recording card. See 1.2/1 for compression pressure.

7 Detach compression pressure tester and special tool No. 17.

8 Install injection nozzle, referring to 07.8—4.1/1.



Survey

Version

Air filter

09.8

Turbocharger 3 LKs

09.13

Turbocharger To 4 B 27

09.14

Contents

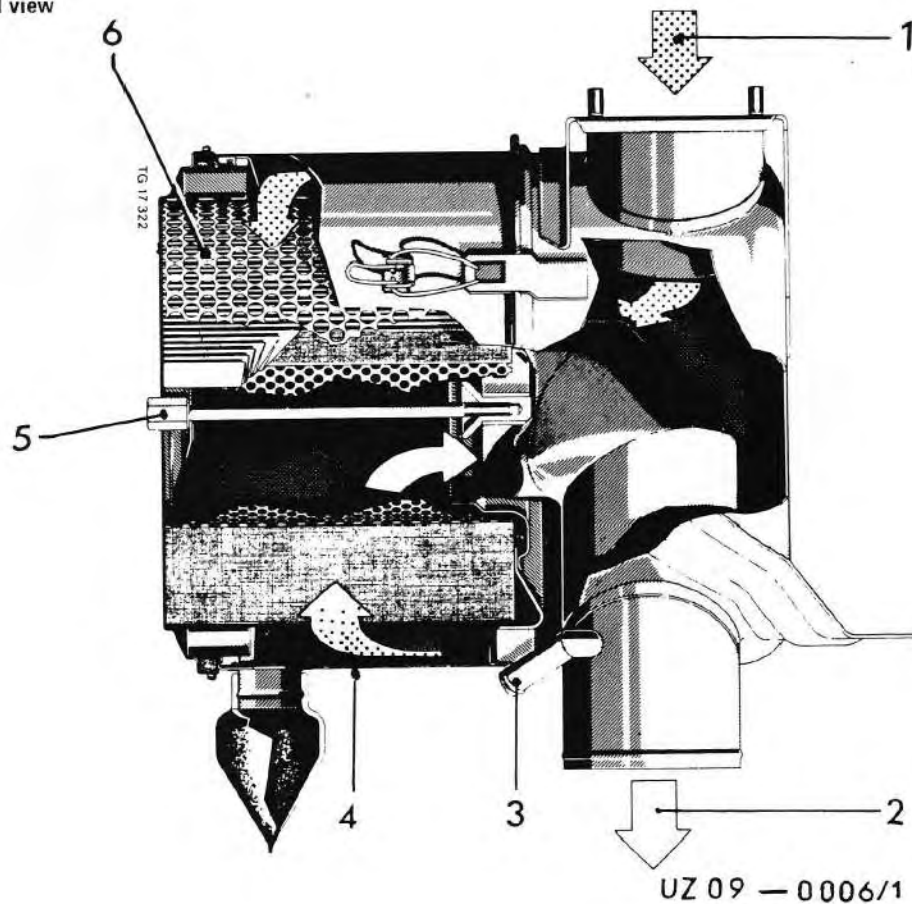
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Installation survey
Air filter

Chassis model		435.110/.111/.113/.115/.117	
Air filter (choice)	Type	Combination air filter	
	Manufacturer	MANN-Piclon	micro-star LVP2505
Filter element (choice)	Type	MANN & Hummel	Knecht
	Manufacturer	C 27 585	micro-star AG 57
	Type	MANN & Hummel	Knecht
	Manufacturer		

Sectional view

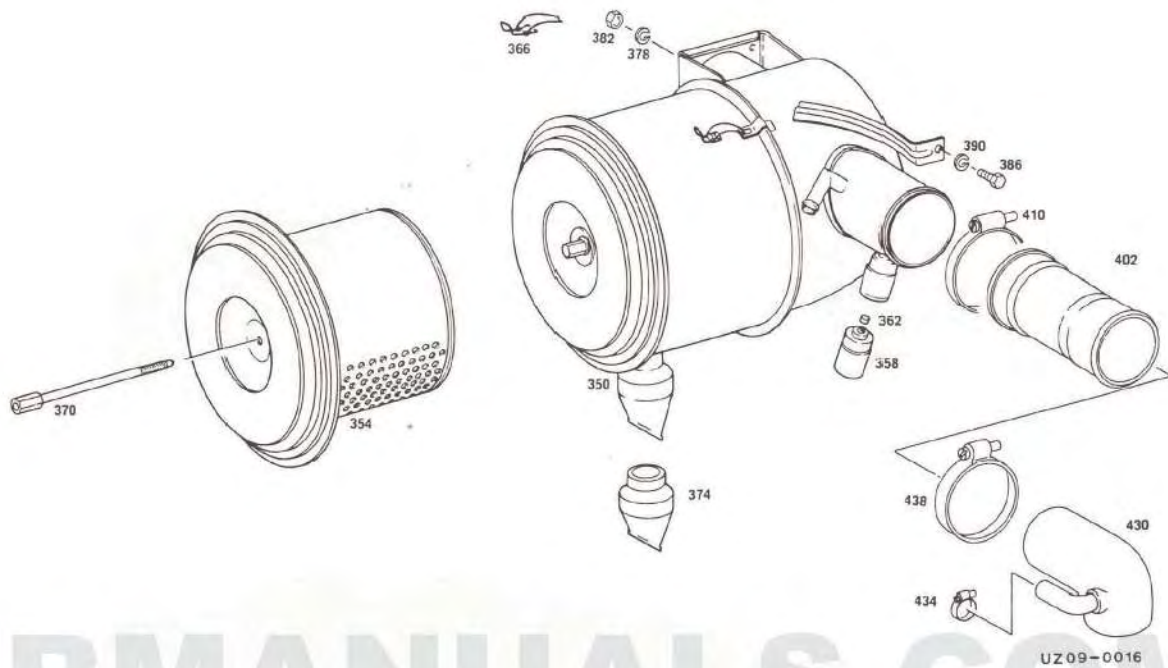


Air filter

- 1 Air inlet
- 2 Air departure
- 3 Connection air compressor

- 4 Filter bearing
- 5 Screw plug
- 6 Filter element with cover

Exploded view



Air filter

- 350 Air filter
- 354 Filter element
- 358 Vacuum unit
- 362 Washer
- 366 Lock
- 370 Bolt
- 374 Valve
- 378 Spring lock washer

- 382 Nut
- 386 Bolt
- 390 Spring lock washer
- 402 Intake pipe
- 410 Clamp
- 430 Manifold
- 434 Clamp
- 438 Clamp